Contextual Stochastic Optimization for Determining Electric Vehicle Charging Station Locations with Decision-Dependent Demand Learning

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Abstract

We consider a two-stage contextual stochastic electric vehicle (EV) charging station location problem, where the uncertain customer demand depends on both exogenous contextual information (e.g., gross domestic product and education level) and our first-stage location/capacity decisions, leading to *decision-dependent uncertainty*. For example, opening a charging station increases the demand around that area. We model this problem using the empirical residuals-based decision-dependent sample average approximation (ER-DD-SAA) framework, which explicitly embeds demand learning into the optimization by leveraging the empirical residuals. We theoretically analyze the consistency and asymptotic optimality of the ER-DD-SAA framework. To learn the latent decision-dependent customer demand, we propose a nonlinear regression model that involves an exponential term on the distance between charging stations and customer sites, which is challenging to regress directly. To overcome this issue, we devise a two-step regression approach. We first regress on the distance and location/capacity decisions while keeping all other exogenous covariates the same. In the second step, we use classical linear regression as well as modern machine learning tools (e.g., random forest, decision trees, and gradient boosting) to learn the impact of the rest of the covariates on customer demand. We conduct synthetic experiments and a case study based on real-world data in New York State to illustrate the effectiveness of our proposed ER-DD-SAA model.

Keywords: Contextual stochastic optimization, Decision-dependent uncertainty, Machine learning, Electric vehicle charging station location problem

1 Introduction

With growing environmental consciousness and government regulations, electric vehicles (EVs) are experiencing a rise in popularity. In 2023, nearly 14 million EVs were registered worldwide, bringing the total to 40 million EVs. In 2024, EV sales increased by 3.5 million from the previous year, marking a 35% rise and more than a sixfold increase compared to 2018 sales, just five years earlier

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(International Energy Agency, 2024). A key barrier to even wider adoption of EVs is the inadequacy of charging infrastructure. Many potential EV drivers are afraid of running out of battery without being able to find a charging station in time to replenish it, leading to the so-called "range anxiety" (EVBOX, 2023). To alleviate such concerns and to enable a more sustainable transportation infrastructure, strategically determining the optimal locations of EV charging stations has become a pressing need. For example, the United States' Bipartisan Infrastructure Law has allocated funds for deploying EV charging infrastructure and set the goal of installing 500,000 EV chargers by 2030 (The White House, 2022). Canada has its Zero-Emission Vehicle Infrastructure Program that aims to achieve 100% zero-emission vehicle sales by 2035 (Government of Canada, 2024). The European Union has an Alternative Fuels Infrastructure Regulation that requires EV chargers every 60 km along major highways by 2026 (European Commission, 2024). All of these initiatives require a gradual rollout of EV charging stations over a period of time while aligning infrastructure development with evolving demand patterns.

In this paper, we consider a two-stage EV charging station location problem to address this pressing decision-making problem. Specifically, in the first stage, we select the best locations from a set of candidate facilities to install public EV charging stations as well as optimize the capacity of each EV charging station (i.e., how many chargers to install in each location). Then, in the second stage, demand is realized based on our deployment decisions and fulfilled by the open charging stations. A unique feature of our model is that it integrates a "learning" part into the optimization model that estimates the demand based on our decisions as well as other relevant side information. We discuss this next.

One aspect that complicates the EV charging station location problem is that the underlying uncertain customer demand can be affected by both the endogenous location/capacity decisions that are being optimized and the exogenous contextual information (also known as side information, covariates, or features). Exogenous factors that can influence customer demand include gross domestic product (GDP) of a region, education levels, gas prices, and tax incentives (Achtnicht et al., 2012; Sierzchula et al., 2014; Langbroek et al., 2016; Li et al., 2017b,a; Ma and Fan, 2020). A higher GDP indicates greater purchasing ability for EV adoption, while higher education levels often correlate with increased environmental awareness and technology acceptance. Gas prices and tax incentives for purchasing EVs are two other major factors that drive consumers to switch to EVs. However, the tax incentives, once announced, are quite stable, which makes it hard to regress its impact on customer demand. On the other hand, the fuel prices are often recorded at a state level from open-source datasets, which do not have the granularity required in our analyses. Due to these reasons, we drop gas prices and tax incentives from the feature space and base our analyses on GDPs and education levels only. We note that the methodologies presented in this paper are quite general and can be applied to incorporate other features as well.

Location and capacity decisions also have an impact on customer demands, indicating that the uncertain customer demand is *decision-dependent*. For example, if more chargers are opened in a neighborhood, people will be more inclined to purchase EVs in that area because they will have less range anxiety (Basciftci et al., 2021; Mahmutoğulları and Yaman, 2023; Liu et al., 2022; Li

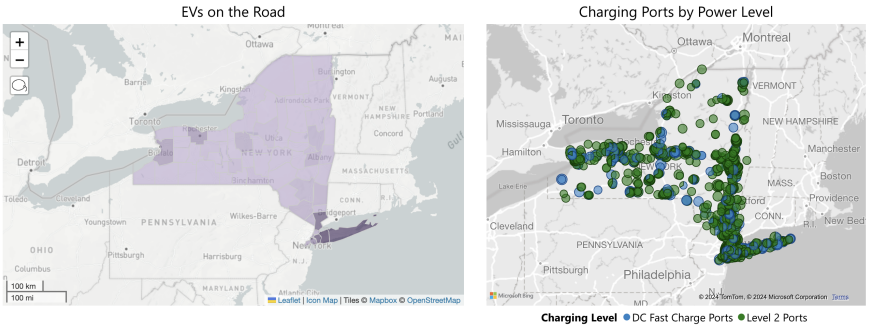


Figure 1: Electric vehicle adoption (left: darker shade indicates higher adoption rate) and charging infrastructure (right) in New York State.

et al., 2024; Deng and Sen, 2022). This intuition can be inferred from the real-world dataset in New York State, as shown in Figure 1 (Center, 2023), where areas with a higher density of EV charging stations tend to have more EVs on the road. This decision dependency makes it difficult to predict demand precisely because it requires costly investment decisions and then observing its effects. To the best of our knowledge, most of the existing literature on incorporating decision-dependent uncertainty in facility location problems (which will be reviewed next) assumed an affine decision dependency form with fixed coefficients and did not use real-world data to estimate the latent decision dependency. To bridge this gap, our paper, for the first time to our knowledge, proposes a general regression model for customer demand that incorporates both endogenous location/capacity decisions that are part of the optimization model and exogenous covariate information. In addition, unlike many existing studies, this work calibrates the proposed regression model using real-world datasets. To account for the estimation errors, we adopt the empirical residuals-based decision-dependent sample average approximation (ER-DD-SAA) framework and theoretically prove its consistency and asymptotic optimality under some mild assumptions.

Next, we review relevant literature on EV charging station location and contextual stochastic optimization in more detail in Section 1.1 and summarize our main contributions in Section 1.2.

1.1 Related Work

This paper contributes to two main lines of research, one on EV charging station location problems and the other on contextual stochastic optimization.

EV Charging Station Location. We refer the readers to Shen et al. (2019) for a comprehensive survey of optimization models and solution approaches for EV charging infrastructure planning, EV charging operations, and public policy/business models. Here, we briefly mention main research works relevant to this study. We begin with *deterministic* models that incorporated the relationship between the EV demand and deployment decisions. Zhang et al. (2017) and Anjos et al. (2020) studied multi-period deterministic EV charging station location problems where they modeled the demand dynamics such that the number of potential new EV adopters depends on the number of EV owners in the previous period as well as charging opportunities. However, they assumed

known demand dynamics without any uncertainty, which can be unrealistic in various real-world applications. Next, we review relevant papers on *stochastic* EV charging station location problems under uncertain demand. Wu and Sioshansi (2017) proposed a two-stage stochastic flow-capturing location model, where the uncertain EV charging demand is modeled as flow on each trip and each EV charging station can capture the flow-based demand within a certain distance. To solve this model, they employed sample average approximation (SAA) framework, generating scenarios from historical tour data to represent the uncertain demand. Çalık and Fortz (2019) formulated a two-stage stochastic programming model for a one-way electric car-sharing system to locate recharging stations and developed a Benders decomposition algorithm to solve it. Li et al. (2021) studied a two-stage distributionally robust mixed-integer linear programming model that incorporated the probability distribution of demand uncertainty through an L_1-norm ambiguity set and reformulated the problem as a convex combination of conditional value-at-risk and the worst-case cost under the reference distribution.

The majority of the existing works model the EV demand as an exogenous uncertainty (independent of decisions). There are only a few papers that optimize facility locations under *decision-dependent demand*. Basciftci et al. (2021) developed a two-stage distributionally robust facility location model with moment-based ambiguity sets, where the first-stage binary location decisions affinely affect the first- and second-moment of the uncertain demand. They assumed a specific functional form of decision dependency with fixed parameters. They then reformulated the model as a monolithic mixed-integer linear program and proposed valid inequalities to enhance solution tractability. Mahmutoğulları and Yaman (2023) studied a robust refueling station location problem with routing decisions, where location decisions affect flow-based demands through a decision-dependent polyhedral uncertainty set, with lower and upper bounds of flow parameters affinely depending on the location decision variables. They derived mixed-integer linear programming reformulations and implemented Benders decomposition within a branch-and-cut framework. Li et al. (2024) formulated a two-stage decision-dependent distributionally robust optimization model for fast charging station planning, where the incremental EV adoption rate is influenced by the convenience level of charging that is modeled as a piecewise linear function of location decisions and exogenous factors such as EV prices and subsidies. They reformulated the problem into a single-level mixed-integer linear program by exploiting strong duality and McCormick envelopes. However, all of these papers assumed an affine decision dependency of the uncertain demand with pre-given parameters and none of them leveraged historical data to simultaneously learn the impact of exogenous and endogenous factors on uncertain customer demand and optimize the location/capacity decisions. In contrast, in this paper, we study a two-stage decision-dependent stochastic EV charging station location problem that integrates an estimated relationship between the uncertain EV demand and both the endogenous location/capacity decisions and the exogenous contextual information, learned from real-world data, into the optimization model.

Contextual Stochastic Optimization. In recent years, there has been a stream of research that integrates prediction models and optimization under uncertainty, called *contextual stochastic*

optimization. We refer interested readers to extensive surveys in Qi and Shen (2022) and Sadana et al. (2025). These methods can be categorized as (i) Smart "predict, then optimize" (SPO) (Elmachtoub and Grigas, 2022; El Balghiti et al., 2019; Estes and Richard, 2023), where a new SPO loss function is proposed that aims to measure the decision error in the downstream optimization problem directly; (ii) Estimate-then-optimize methods (Bertsimas and Kallus, 2020; Bertsimas et al., 2023) that incorporate learning of the conditional distributions into the optimization step; (iii) Feature-to-decision learning (Ban and Rudin, 2019; Bertsimas and Koduri, 2022; Zhang et al., 2024; Qi et al., 2024) that seeks to learn the optimal feature-to-decision mapping directly; and (iv) Empirical residuals-based sample average approximation (ER-SAA) (Ban et al., 2019; Kannan et al., 2025; Deng and Sen, 2022; Yurdakul and Bayraksan, 2024) and its distributionally robust optimization variant (ER-DRO) (Kannan et al., 2024; Qi et al., 2022) that incorporate the empirical residuals obtained from the regression step into the optimization model to account for estimation error. However, most of these works did not incorporate decision-dependent uncertainty, with only a few exceptions. Bertsimas and Kallus (2020) leveraged various local-learning methods to assign weights to each historical data point, creating a so-called reweighted SAA. When decisions also affect the uncertainty, they adapted the learning methods to estimate weights that depend on both endogenous decisions and exogenous covariates. Liu et al. (2022) proposed the coupled learning enabled optimization (CLEO) algorithm that couples local regression models and trust region methods for solving stochastic programming problems with decision-dependent uncertainty. The CLEO algorithm iteratively estimates the latent decision dependency and updates a candidate decision, achieving computational tractability while guaranteeing almost sure convergence to a stationary point. More recently, Zhu et al. (2024) extended the ER-DRO approach to handle decision-dependent uncertainty in a contextual setting and derived statistical guarantees for various ambiguity sets formed via the Wasserstein distance, a sample robust approach, and phi-divergences.

Our paper focuses on the ER-SAA approach instead (as opposed to its DRO variant) and extends it to the decision-dependent setting. We refer to this method as ER-DD-SAA. Specifically, we first utilize historical covariate information and decisions to estimate the latent decision-dependent uncertainty via a two-step regression approach. The empirical residuals from training the predictive model are then incorporated to construct different scenarios around the point estimators within an SAA framework. For the contextual stochastic EV charging station location problem under decision-dependent demand uncertainty, we show that the ER-DD-SAA asymptotically converges to the true problem under some mild assumptions. While a DRO variant has recently been investigated Zhu et al. (2024), to the best of our knowledge, this work is the first to examine ER-DD-SAA framework and apply a contextual stochastic model to the EV charging station location problem.

We end this section by pointing out that in the recent survey by Sadana et al. (2025), the authors identified contextual stochastic optimization with decision-dependent uncertainty as a future research direction. As mentioned in their review, "while there has been some progress in studying problems where the decision affects the uncertain parameters, the literature on decision-dependent uncertainty with covariates is sparse". This paper bridges this gap by studying a contextual EV charging station location problem under decision-dependent uncertainty.

1.2 Contributions and Outline

To summarize, our key contributions are as follows:

(i) We consider a two-stage contextual EV charging station location problem under decision-dependent demand. Our demand model incorporates both endogenous location/capacity decisions and exogenous covariate information such as GDPs and education levels. To the best of our knowledge, this is the first study to integrate decision dependency and contextual information in an EV charging station location problem.

(ii) We extend the ER-SAA framework proposed in Kannan et al. (2025) to the decision-dependent setting, namely, ER-DD-SAA. We theoretically show that the optimal solutions and objective values of the ER-DD-SAA problem asymptotically converge to the ones of the true problem under mild assumptions.

(iii) We adopt a general nonlinear regression model to estimate EV customer demands, which is challenging to regress directly. For computational tractability, we introduce a novel two-step regression approach that can incorporate both linear and nonlinear, nonparametric regression models for approximately learning decision-dependent demand.

(iv) To test our proposed two-step regression approach, we use synthetic datasets to estimate the coefficients via the proposed two-step regression approach and compare them with the ones obtained from one-step joint regression. Our numerical results demonstrate that the proposed two-step approach can solve all instances within one second while maintaining good predictive power, whereas the one-step regression becomes computationally intractable as the data size increases. We also conduct sensitivity analyses by varying different problem parameters, including the degree of decision dependency, and compare our ER-DD-SAA with the decision-independent counterpart (ER-DI-SAA).

(v) Finally, we validate our methodology using a real-world dataset from the state of New York and conduct numerical experiments to demonstrate its effectiveness.

The subsequent sections of this paper are structured as follows. In Section 2, we introduce the two-stage contextual stochastic EV charging station location problem with decision-dependent demand. We propose the ER-DD-SAA framework and theoretically prove its consistency and asymptotic optimality. In Section 3, to learn the decision-dependent demand, we propose both parametric and nonparametric models and employ a two-step regression analysis to determine the model coefficients. To better conduct the out-of-sample experiments and sensitivity analyses, we first test our model in synthetic datasets with varying problem parameters in Section 4. We also conduct a case study using a real-world dataset in New York State to illustrate the effectiveness of the proposed demand learning and ER-DD-SAA model in Section 5. In both numerical studies, we compare our contextual decision-dependent model with the contextual decision-independent counterpart to illustrate the importance of considering decision dependency.

2 Contextual Stochastic EV Charging Station Location Problem with Decision-Dependent Demand Learning

In our EV charging station location problem, we assume that the EV customer demand is influenced by exogenous factors such as the distance between charging stations and customer sites, education level, and GDP, as well as endogenous factors such as location/capacity decisions. We first formulate the two-stage contextual stochastic EV charging station location problem in Section 2.1. To address the estimation error in the regression step, we propose an ER-DD-SAA-based formulation tailored for our model in Section 2.2 and analyze its theoretical properties in Section 2.3.

2.1 Two-Stage Contextual Stochastic EV Charging Station Location Problem

Consider a set of candidate facilities $i \in I$ to locate EV charging stations and a set of customer sites $j \in J$, each representing a geographical area (e.g., a county). Define first-stage binary variables $y_i \in \{0,1\},\ i \in I$ as the location decisions, such that $y_i = 1$ if we open a charging station at location $i \in I$ and $y_i = 0$ otherwise. Define first-stage integer variables $z_i \in \mathbb{Z},\ i \in I$ as the number of chargers installed at location i, with an upper limit M_i for each location $i \in I$. Continuous variables x_{ij} and s_j represent the amount of demand at customer site $j \in J$ fulfilled by charging station i and the unsatisfied demand at customer site j for all $i \in I,\ j \in J$, respectively. Let dist_{ij} represent the distance between customer site $j \in J$ and location $i \in I$, and let g_j and e_j denote the GDP and education level at customer site $j \in J$, respectively. As mentioned, our model can be easily adapted to use other covariate information based on data availability. For ease of exposition and to match our real-world dataset, we continue with these covariates. We adopt bold notation for vector forms. Given first-stage decision vector $\boldsymbol{z}$ and covariate information $\boldsymbol{w} = (\mathbf{dist}, \mathbf{g}, \mathbf{e}) \in \mathcal{W}$, the random EV demand at customer site $j \in J$ is denoted as $d_j(\boldsymbol{z}, \boldsymbol{w})$. We will introduce specific ways to model this demand function in Section 3. The function $h\big(\boldsymbol{z}, \boldsymbol{d}(\boldsymbol{z}, \boldsymbol{w})\big)$ represents the second-stage cost given first-stage decision $\boldsymbol{z}$ and corresponding customer demand $\boldsymbol{d}(\boldsymbol{z}, \boldsymbol{w})$. Parameters $f_i, b_i, c_{ij}, p_j, r_j, C_i$ denote the fixed cost for opening station $i \in I$, unit installation cost for installing one charger at station $i \in I$, unit transportation cost from station $i \in I$ to customer site $j \in J$, unit penalty for unsatisfied demand at customer site $j \in J$, unit revenue for demand fulfillment at site $j \in J$, and the maximum number of EV demand each charger can fulfill at location $i \in I$, respectively.

With the above notation, given a new observation $\boldsymbol{W} = \boldsymbol{w}$, our two-stage contextual stochastic EV charging station location problem is formulated as follows:

$$\min_{\boldsymbol{y},\boldsymbol{z}} \quad \sum_{i \in I} \Big(f_i y_i + b_i z_i \Big) + \mathbb{E}\Big[h\big(\boldsymbol{z}, \boldsymbol{d}(\boldsymbol{z}, \boldsymbol{w})\big) \;\big|\; \boldsymbol{Z} = \boldsymbol{z}, \boldsymbol{W} = \boldsymbol{w} \Big] \tag{1a}$$

$$\text{s.t.} \quad z_i \le M_i y_i, \quad \forall i \in I, \tag{1b}$$

$$y_i \in \{0,1\},\ z_i \in \mathbb{Z}, \quad \forall i \in I, \tag{1c}$$

where

$$h\big(\boldsymbol{z}, \boldsymbol{d}(\boldsymbol{z}, \boldsymbol{w})\big) = \min_{\boldsymbol{x}, \boldsymbol{s}} \left(\sum_{i \in I} \sum_{j \in J} c_{ij} x_{ij} + \sum_{j \in J} \Big(p_j s_j - r_j d_j(\boldsymbol{z}, \boldsymbol{w}) \Big) \right) \tag{2a}$$

$$\text{s.t.} \sum_{i \in I} x_{ij} + s_j = d_j(\boldsymbol{z}, \boldsymbol{w}), \quad \forall j \in J, \tag{2b}$$

$$\sum_{j \in J} x_{ij} \leq C_i z_i, \quad \forall i \in I, \tag{2c}$$

$$x_{ij}, s_j \geq 0, \quad \forall i \in I,\ j \in J. \tag{2d}$$

The objective function (1a) aims to minimize the total investment cost of opening EV charging stations and installing chargers as well as the expected second-stage operational cost, conditioned on the observed contextual information $\boldsymbol{w}$ and the capacity decisions $\boldsymbol{z}$. Constraints (1b) ensure that chargers can only be installed if a charging station is opened at location $i \in I$ and the number of chargers installed at location $i \in I$ cannot exceed the upper limit M_i. Given first-stage decisions $\boldsymbol{z}$ and customer demand $\boldsymbol{d}(\boldsymbol{z}, \boldsymbol{w})$, the objective function (2a) in the second-stage problem minimizes the total transportation cost and penalty on the unmet demand minus the total revenue. Constraints (2b) ensure that each demand at customer site $j \in J$ is either fulfilled by a charging station or recorded as unmet demand. Additionally, constraints (2c) impose capacity limitations by restricting the total fulfilled demand at each open station $i \in I$ to be no more than the number of chargers z_i times the charging capacity per charger C_i. Lastly, constraints (2d) ensure the non-negativity of the satisfied and unmet demand.

2.2 ER-DD-SAA

To present the proposed ER-DD-SAA framework to solve the contextual decision-dependent stochastic EV charging statation location problem and to facilitate the theoretical analysis of our ER-DD-SAA framework, we first reformulate Model (1) in the following matrix form:

$$\min_{\boldsymbol{y}, \boldsymbol{z} \in \mathcal{Z}} \mathbb{E}_{\boldsymbol{d}}\big[c(\boldsymbol{y}, \boldsymbol{z}; \boldsymbol{d}(\boldsymbol{z}, \boldsymbol{w})) | \boldsymbol{Z} = \boldsymbol{z}, \boldsymbol{W} = \boldsymbol{w}\big] := \mathbb{E}_{\boldsymbol{d}}\big[\boldsymbol{f}^T \boldsymbol{y} + \boldsymbol{b}^T \boldsymbol{z} + h\big(\boldsymbol{z}, \boldsymbol{d}(\boldsymbol{z}, \boldsymbol{w})\big) | \boldsymbol{Z} = \boldsymbol{z}, \boldsymbol{W} = \boldsymbol{w}\big], \tag{3}$$

where $\mathcal{Z} := \{z_i \leq M_i y_i,\ y_i \in \{0,1\},\ z_i \in \mathbb{Z},\ \forall i \in I\}$ is the compact first-stage feasible region. Here, the second-stage cost $h\big(\boldsymbol{z}, \boldsymbol{d}(\boldsymbol{z}, \boldsymbol{w})\big)$ can be written as the following minimization problem with feasible set $\mathcal{H}(\boldsymbol{z}, \boldsymbol{w})$ defined by constraints (2b)–(2d):

$$h\big(\boldsymbol{z}, \boldsymbol{d}(\boldsymbol{z}, \boldsymbol{w})\big) = \min_{\boldsymbol{x}, \boldsymbol{s} \in \mathcal{H}(\boldsymbol{z}, \boldsymbol{w})} \boldsymbol{c}^T \boldsymbol{x} + \boldsymbol{p}^T \boldsymbol{s} - \boldsymbol{r}^T \boldsymbol{d}(\boldsymbol{z}, \boldsymbol{w}). \tag{4}$$

We assume that the true relationship between the customer demand, capacity decisions, and covariate information can be represented as

$$\boldsymbol{d}(\boldsymbol{Z}, \boldsymbol{W}) = Q^*(\boldsymbol{Z}, \boldsymbol{W}) + \boldsymbol{\epsilon}, \tag{5}$$

where $Q^*(\boldsymbol{z}, \boldsymbol{w}) := \mathbb{E}[\boldsymbol{d}(\boldsymbol{Z}, \boldsymbol{W})|\boldsymbol{Z} = \boldsymbol{z}, \boldsymbol{W} = \boldsymbol{w}]$ is the (unknown) ground-truth regression function, and $\boldsymbol{\epsilon}$ represents the zero-mean error term. Later in Section 3.1, we will present a specific demand model for our problem. The true conditional stochastic program (CSP) to our EV charging station location problem (3) is then equivalent to

$$v^*(\boldsymbol{w}) := \min_{\boldsymbol{y}, \boldsymbol{z} \in \mathcal{Z}} \left\{ g(\boldsymbol{y}, \boldsymbol{z}; \boldsymbol{w}) := \mathbb{E}_{\boldsymbol{\epsilon}}[c(\boldsymbol{y}, \boldsymbol{z}; Q^*(\boldsymbol{z}, \boldsymbol{w}) + \boldsymbol{\epsilon})] \right\}, \tag{6}$$

where the expectation is taken with respect to the distribution of $\boldsymbol{\epsilon}$. We refer to (6) as the "true" problem and denote its optimal solution as $\boldsymbol{y}^*(\boldsymbol{w}), \boldsymbol{z}^*(\boldsymbol{w})$ and optimal solution set as $S^*(\boldsymbol{w})$.

In practice, it is not possible to compute the expectation that appears in (6) because (i) we may not know the true distribution, and (ii) even if we know the distribution, this involves a very complicated random variable given as the objective function value of a two-stage contextual stochastic optimization problem. Instead, in real life, we may have access to a dataset $\mathcal{D}_n = \{(\boldsymbol{z}^i, \boldsymbol{w}^i, \boldsymbol{d}^i)\}_{i=1}^n$ from historical observations. We will soon present a way to approximate the true problem (6) by a data-driven approximation using the dataset $\mathcal{D}_n$. Before we do that, we first present an intermediate problem that will be used in the subsequent analysis.

Denote $\boldsymbol{\epsilon}^i := \boldsymbol{d}^i - Q^*(\boldsymbol{z}^i, \boldsymbol{w}^i)$ for $i = 1, \ldots, n$ as the "true" residuals. If the true regression function $Q^*(\boldsymbol{z}, \boldsymbol{w})$ is known, we can construct the full-information decision-dependent SAA (FI-DD-SAA) as follows:

$$v_n^*(\boldsymbol{w}) := \min_{\boldsymbol{y}, \boldsymbol{z} \in \mathcal{Z}} \left\{ g_n^*(\boldsymbol{y}, \boldsymbol{z}; \boldsymbol{w}) := \frac{1}{n} \sum_{i=1}^n c(\boldsymbol{y}, \boldsymbol{z}; Q^*(\boldsymbol{z}, \boldsymbol{w}) + \boldsymbol{\epsilon}^i) \right\}. \tag{7}$$

However, the true regression function $Q^*(\boldsymbol{z}, \boldsymbol{w})$ is rarely known in real-world applications. An alternative is to first estimate $Q^*(\boldsymbol{z}, \boldsymbol{w})$ from the dataset $\mathcal{D}_n$, e.g., by minimizing some loss function.

Denoting the regression estimator that uses the n observations as $\hat{Q}_n(\boldsymbol{z}, \boldsymbol{w})$, we compute the empirical residuals as $\hat{\boldsymbol{\epsilon}}^i = \boldsymbol{d}^i - \hat{Q}_n(\boldsymbol{z}^i, \boldsymbol{w}^i)$ for $i = 1, \ldots, n$, which serve as an approximation of the samples of $\boldsymbol{\epsilon}$. Adding these empirical residuals to the point estimator $\hat{Q}_n(\boldsymbol{z}, \boldsymbol{w})$, we construct the corresponding ER-DD-SAA to the true problem (6) as follows

$$\hat{v}_n^{ER}(\boldsymbol{w}) := \min_{\boldsymbol{y}, \boldsymbol{z} \in \mathcal{Z}} \left\{ \hat{g}_n^{ER}(\boldsymbol{y}, \boldsymbol{z}; \boldsymbol{w}) := \frac{1}{n} \sum_{i=1}^n c(\boldsymbol{y}, \boldsymbol{z}; \hat{Q}_n(\boldsymbol{z}, \boldsymbol{w}) + \hat{\boldsymbol{\epsilon}}^i) \right\}. \tag{8}$$

The above ER-DD-SAA problem, unlike (6), can be constructed using the available data $\mathcal{D}_n$ without needing to calculate the true expectation. We denote its optimal solution as $\hat{\boldsymbol{y}}_n^{ER}(\boldsymbol{w}), \hat{\boldsymbol{z}}_n^{ER}(\boldsymbol{w})$ with optimal solution set $\hat{S}_n^{ER}(\boldsymbol{w})$. These can be viewed as an approximation of the their true counterparts. Next, we analyze the theoretical properties of the ER-DD-SAA problem (8).

2.3 Theoretical Properties of ER-DD-SAA

Under some mild assumptions on the data-generating process and the regression estimates, we show that the optimal value of the ER-DD-SAA problem (8) asymptotically converges to that of

true problem (6). We show similar desirable results for the set of optimal solutions. All proofs are shown in Appendix A. In the subsequent analysis, all norms denoted by $\|\cdot\|$ are Euclidean norms, and $\xrightarrow{P}$ represents convergence in probability. For two sets $S_1, S_2 \subseteq \mathbb{R}^{d_z}$, the deviation of S_1 from S_2 is defined to be $\mathbb{D}(S_1, S_2) := \sup_{a\in S_1} \operatorname{dist}(a, S_2)$, where $\operatorname{dist}(a, S_2) := \inf_{b\in S_2} \|a - b\|$.

We begin by showing that the cost function in (3) is Lipschitz continuous with respect to $\boldsymbol{d}$.

Proposition 1. For each $\boldsymbol{y}, \boldsymbol{z} \in \mathcal{Z}$, the cost function c in problem (6) satisfies Lipschitz condition

$$\left|c(\boldsymbol{y}, \boldsymbol{z}; \bar{\boldsymbol{d}}) - c(\boldsymbol{y}, \boldsymbol{z}; \boldsymbol{d})\right| \leq L\|\bar{\boldsymbol{d}} - \boldsymbol{d}\|, \ \forall \bar{\boldsymbol{d}}, \boldsymbol{d} \in \mathbb{R}^{|J|}_{\geq 0}$$

with Lipschitz constant $L = \max_{\boldsymbol{u},\boldsymbol{\gamma}\in \operatorname{vert}(\mathcal{F})} \|\boldsymbol{u} - \boldsymbol{r}\| < +\infty$, where $\mathcal{F}$ is the dual feasible set for the second-stage problem (4), $\operatorname{vert}(\mathcal{F})$ is the finite set of extreme points of $\mathcal{F}$, and $\boldsymbol{u}, \boldsymbol{\gamma}$ are dual variables.

The following assumption is useful to derive the consistency of the objective function $g_n^*(\boldsymbol{y}, \boldsymbol{z}; \boldsymbol{w})$.

Assumption 1. (i) The weak Law of Large Numbers (LLN) holds pointwise for error samples $\{\boldsymbol{\epsilon}^i\}_{i=1}^n$; (ii) $c(\boldsymbol{y}, \boldsymbol{z}; \boldsymbol{d}(\boldsymbol{z}, \boldsymbol{w}))$ is dominated by an integrable function.

Assumption 1(i) is naturally satisfied when the data $\{\boldsymbol{\epsilon}^i\}_{i=1}^n$ is independent and identically distributed (i.i.d.). However, in some real-world settings, the data may not be i.i.d. For such settings, Assumption 1(i) holds for a variety of mixing/stationary processes; see, e.g., McLeish et al. (1975). Assumption 1(ii) holds if $\mathbb{E}\big[|\boldsymbol{d}(\boldsymbol{z}, \boldsymbol{w})|\big] < +\infty$, which is a mild and reasonable assumption in our model.

Proposition 2. If Assumption 1 holds, then for almost everywhere (a.e.) $\boldsymbol{w} \in \mathcal{W}$, the sequence of sample average $g_n^*(\cdot, \cdot; \boldsymbol{w})$ converges in probability to the true objective function $g(\cdot, \cdot; \boldsymbol{w})$ uniformly on set $\mathcal{Z}$.

The first-stage feasible set $\mathcal{Z}$ is compact by virtue of being discrete and finite. If Assumption 1 holds and the error samples $\{\boldsymbol{\epsilon}^i\}_{i=1}^n$ are i.i.d, then following Theorem 7.48 of Shapiro et al. (2021), the convergence result in Proposition 2 is derived. For non i.i.d. $\{\boldsymbol{\epsilon}^i\}_{i=1}^n$ satisfying Assumption 1, the same convergence result holds by adapting the proof of Theorem 7.48 in Shapiro et al. (2021) to use weak LLN of $\{\boldsymbol{\epsilon}^i\}_{i=1}^n$. Therefore, we omit the proof of Proposition 2.

Recall that the feasible region of the EV charging station location problem is given by $\mathcal{Z} := \{z_i \leq M_i y_i, \ y_i \in \{0, 1\}, \ z_i \in \mathbb{Z}, \ \forall i \in I\}$, where the decisions $\boldsymbol{z}$ denote the number of chargers installed in each location. Let $\mathcal{Z}_z := \prod_{i=1}^{|I|}\{0, 1, \ldots, M_i\}$ be a superset of the domain of the $\boldsymbol{z}$-variable.

Assumption 2. We assume that the regression estimate $\hat{Q}_n(\boldsymbol{z}, \boldsymbol{w})$ has the following consistency properties:

(i) For a.e. $\boldsymbol{w} \in \mathcal{W}$, $\hat{Q}_n(\boldsymbol{z}, \boldsymbol{w}) \xrightarrow{P} Q^*(\boldsymbol{z}, \boldsymbol{w})$ uniformly on $\mathcal{Z}_z$.

(ii) $\frac{1}{n}\sum_{i=1}^{n} \|Q^*(\boldsymbol{z}^i, \boldsymbol{w}^i) - \hat{Q}_n(\boldsymbol{z}^i, \boldsymbol{w}^i)\| \xrightarrow{P} 0$.

Note that the uniform convergence in Assumption 2(i) is stronger than the pointwise convergence required in the decision-independent setting (see Assumption (5a) in Kannan et al. (2025)). However, given that we have a compact superset $\mathcal{Z}_z$, it can be satisfied by many regression models if the regression estimators are weakly consistent. To see this, take our demand model (9) and assume $f_2(\mathrm{g}_j, \mathrm{e}_j)$ to be linear in $\mathrm{g}_j, \mathrm{e}_j$, i.e., $Q^*(\boldsymbol{z}, \boldsymbol{w}) = \sum_{i\in I} \min\{t_i, T\} e^{\alpha_1^* - \beta_1^* \mathbf{dist}_i} z_i + \beta_2^* \mathbf{g} + \beta_3^* \mathbf{e} + \boldsymbol{\alpha}_2^*$. Let $\hat{\alpha}_{1,n}, \hat{\beta}_{1,n}, \hat{\beta}_{2,n}, \hat{\beta}_{3,n}, \hat{\boldsymbol{\alpha}}_{2,n}$ denote the corresponding estimated coefficients in $\hat{Q}_n(\boldsymbol{z}, \boldsymbol{w})$. Then, we have for a.e. $\boldsymbol{w} \in \mathcal{W}$

$$
\begin{aligned}
&\sup_{\boldsymbol{z}\in\mathcal{Z}_z} \left\| \hat{Q}_n(\boldsymbol{z}, \boldsymbol{w}) - Q^*(\boldsymbol{z}, \boldsymbol{w}) \right\| \\
&= \sup_{\boldsymbol{z}\in\mathcal{Z}_z} \left\| \sum_{i\in I} \min\{t_i, T\} (e^{\hat{\alpha}_{1,n} - \hat{\beta}_{1,n} \mathbf{dist}_i} - e^{\alpha_1^* - \beta_1^* \mathbf{dist}_i}) z_i + (\hat{\beta}_{2,n} - \beta_2^*)\mathbf{g} + (\hat{\beta}_{3,n} - \beta_3^*)\mathbf{e} + (\hat{\boldsymbol{\alpha}}_{2,n} - \boldsymbol{\alpha}_2^*) \right\| \\
&\leq \sum_{i\in I} \min\{t_i, T\} \| e^{\hat{\alpha}_{1,n} - \hat{\beta}_{1,n} \mathbf{dist}_i} - e^{\alpha_1^* - \beta_1^* \mathbf{dist}_i} \| M_i + |\hat{\beta}_{2,n} - \beta_2^*| \|\mathbf{g}\| + |\hat{\beta}_{3,n} - \beta_3^*| \|\mathbf{e}\| + \|\hat{\boldsymbol{\alpha}}_{2,n} - \boldsymbol{\alpha}_2^*\| \\
&\xrightarrow{P} 0,
\end{aligned}
$$

where the inequality is due to $z_i \leq M_i$ for all $i \in I$ and the last line is due to the continuity of the exponential function and the weak consistency of the estimators, i.e., $\hat{\alpha}_{1,n} \xrightarrow{P} \alpha_1^*$, $\hat{\alpha}_{2,n} \xrightarrow{P} \alpha_2^*$, $\hat{\beta}_{1,n} \xrightarrow{P} \beta_1^*$, $\hat{\beta}_{2,n} \xrightarrow{P} \beta_2^*$, $\hat{\beta}_{3,n} \xrightarrow{P} \beta_3^*$ as $n \to +\infty$. Assumption 2(i) is critical for establishing the consistency of the objective function in the ER-DD-SAA problem (see Lemma 3 in Appendix A). On the other hand, Assumption 2(ii) holds if, in addition, the weak LLN of data $\{(\boldsymbol{z}^i, \boldsymbol{w}^i)\}_{i=1}^n$ is satisfied.

We are now ready to present our main result on the desirable asymptotic properties of the proposed ER-DD-SAA framework.

Theorem 1. Suppose Assumptions 1 and 2 hold. Then for the ER-DD-SAA problem (8), for a.e. $\boldsymbol{w} \in \mathcal{W}$, we have

(i) $\hat{v}_n^{ER}(\boldsymbol{w}) \xrightarrow{P} v^*(\boldsymbol{w})$, i.e., the optimal solution for the ER-DD-SAA problem (8) converges in probability to the optimal solution of the true problem (6),

(ii) $\mathbb{D}(\hat{S}_n^{ER}(\boldsymbol{w}), S^*(\boldsymbol{w})) \xrightarrow{P} 0$, i.e., the distance between the optimal solution set of the ER-DD-SAA problem (8) and the optimal solution set of the true problem (6) converges in probability to zero,

(iii) $\sup_{(\boldsymbol{y},\boldsymbol{z})\in\hat{S}_n^{ER}(\boldsymbol{w})} g(\boldsymbol{y}, \boldsymbol{z}; \boldsymbol{w}) \xrightarrow{P} v^*(\boldsymbol{w})$, i.e., the objective value of the true problem (6) with the optimal solution of the ER-DD-SAA problem (8) converges to the true optimal objective value in probability.

3 Decision-Dependent Demand Learning via Two-step Regression

Accurately estimating the customer demand $\boldsymbol{d}(\boldsymbol{z}; \boldsymbol{w})$ is crucial for obtaining high-quality solutions to the EV charging station location problem (1). We present a specific nonlinear decision-dependent

demand model in Section 3.1. To efficiently estimate this nonlinear model, we propose a two-step regression approach, as detailed in Sections 3.2 and 3.3. We then incorporate the empirical residuals obtained from the regression step to the downstream optimization model in Section 3.4.

3.1 Nonlinear Demand Model

To capture the complex relationships between the uncertain customer demand and the exogenous and endogenous factors, we propose a nonlinear regression model in this section. Following Basciftci et al. (2021) and Mahmutoğulları and Yaman (2023), we assume that the customer demand affinely depends on the first-stage capacity decisions $\boldsymbol{z}$ (endogenous factors). Furthermore, we use a general function $f_2(\mathrm{g}_j, \mathrm{e}_j)$ to quantify the impact of exogenous factors (GDPs and education levels) on customer demand, which could be linear or nonlinear nonparametric, as we study in Section 3.3. At any given point of time, we assume demand at customer site $j \in J$ follows

$$d_j(\boldsymbol{z}, \boldsymbol{w}) = \bar{d}_j + \underbrace{\sum_{i \in I} \min\{t_i, T\} e^{\alpha_1 - \beta_1 \mathrm{dist}_{ij}} z_i}_{\text{endogenous component}} + \underbrace{f_2(\mathrm{g}_j, \mathrm{e}_j)}_{\text{exogenous component}} + \epsilon, \quad \forall j \in J, \tag{9}$$

where we explain each component below:

- $\bar{d}_j$ is the initial demand at customer site $j \in J$ (e.g., the demand at the beginning of a year). Note that this value remains constant across different time stamps within a year and can be canceled during our Step-1 regression (see Section 3.2).
- $\min\{t_i, T\}$ is the minimum of t_i, duration between opening of facility $i \in I$ and the opening of next facility, and a threshold T. To account for the diminishing marginal utility effect of opening one more charging station (Marshall, 2013), we assume that the impact of a newly opened station is proportional to $\min\{t_i, T\}$. Note that this assumption is also needed to conduct Step-1 regression as we illustrate in Section 3.2.
- $e^{\alpha_1 - \beta_1 \mathrm{dist}_{ij}}$ is an exponential decay function to account for the impact of distance between charging station $i \in I$ and customer site $j \in J$ on the customer demand (e.g., Basciftci et al., 2021). Under this setting, a closer charging station has a higher impact on customer demand, and this impact decays exponentially in the distance to the charging station.
- $z_i,\ i \in I$ is the endogenous factor representing the number of chargers installed at EV charging station location $i \in I$ during the specified time period.
- $f_2(\mathrm{g}_j, \mathrm{e}_j)$ represents the demand contribution as a function of exogenous factors GDP (g_j) and education level (e_j). We will consider both linear regression and nonlinear nonparametric regression models to describe $f_2(\mathrm{g}_j, \mathrm{e}_j)$ in Section 3.3.
- ϵ denotes the zero-mean noise term. Note that no assumption on normality is needed; however, we assume ϵ does not depend on customer sites $j \in J$.

Since our regression model (9) is nonlinear and the potential charging station locations differ between the training and testing datasets, it is not practical to perform a joint regression on $\boldsymbol{z}, \mathbf{dist}, \mathbf{g}, \mathbf{e}$ using least squares all at once. Therefore, we propose a two-step regression approach as illustrated in Figure 2 to estimate the parameters in the nonlinear regression model (9). Specifically, in the first step, we estimate the coefficients α_1 and β_1 for covariates dist_{ij} and z_i while keeping all other covariates the same, as detailed in Section 3.2. In the second step, we estimate function $f_2(\text{g}_j, \text{e}_j)$ using both linear regression and nonparametric regression approaches with fixed parameters $\hat{\alpha}_1, \hat{\beta}_1$ obtained from the first step, as detailed in Section 3.3. The regression parameters and functions estimated by this two-step process may not be optimal (e.g., in minimizing the mean squared error), but they achieved a good predictive behavior as indicated by our numerical results in Sections 4 and 5. Figure 2 outlines the step-by-step approximation process using the two-step regression analysis together with ER-DD-SAA framework for solving our two-stage stochastic contextual EV charging location problem (1). Let us now discuss each step in more detail.

3.2 Step-1 Regression

We first analyze the demand increase due to the endogenous factors (i.e., the opening of charging stations and their capacity, where a zero capacity indicates facility is not open) to estimate α_1 and β_1. Specifically, in this step, we focus on time-series data that is within one year so that the changes in GDP and education level are negligible. Note that real-world data on these factors are available on an annual basis in our case study.

Table 1 provides an illustrative example that shows the customer demand $d_j(\boldsymbol{z}, \boldsymbol{w})$ at different time stamps, indicating how EV demand at customer site $j \in J$ evolves over time as new stations are added according to our proposed regression model (9). Note that all of these time stamps are within one year so that the GDPs (g_j) and education levels (e_j) remain the same according to public data sources. In this example, we assume that two chargers were installed at location 1 on day D_1 (before which there were no chargers), five chargers were installed at location 2 on day D_2, and eight chargers were installed at location 3 on day D_3. If there were chargers installed in multiple charging stations on the same day, we simply drop this row from the training dataset so that the $\boldsymbol{z}$-matrix is always lower diagonal. On day D_1, two chargers were newly installed at location 1, and their impact on customer demand has yet to begin. As a result, the average demand on day D_1 is $\bar{d}_j + f_2(\text{g}_j, \text{e}_j)$, contributed solely by exogenous factors, according to Model (9). On day D_2, five chargers were newly installed at location 2 and the two chargers at location 1 have been there for $D_2 - D_1$ days. According to Model (9), the average demand on day D_2 is $\bar{d}_j + 2 \times \min\{D_2 - D_1, T\} \times e^{\alpha_1 - \beta_1 \text{dist}_{1j}} + f_2(\text{g}_j, \text{e}_j)$. When the eight chargers were installed at location 3 on day D_3, the effect of location 1's chargers remained the same ($2 \times \min\{D_2 - D_1, T\} \times e^{\alpha_1 - \beta_1 \text{dist}_{1j}}$), and we have an additional term contributed by location 2's chargers ($5 \times \min\{D_3 - D_2, T\} \times e^{\alpha_1 - \beta_1 \text{dist}_{2j}}$). This allows us to subtract the demand for one line from the next line, enabling an effective linear regression on dist_{ij}.

Specifically, the demand increase at customer site $j \in J$ after installing z_i chargers at location

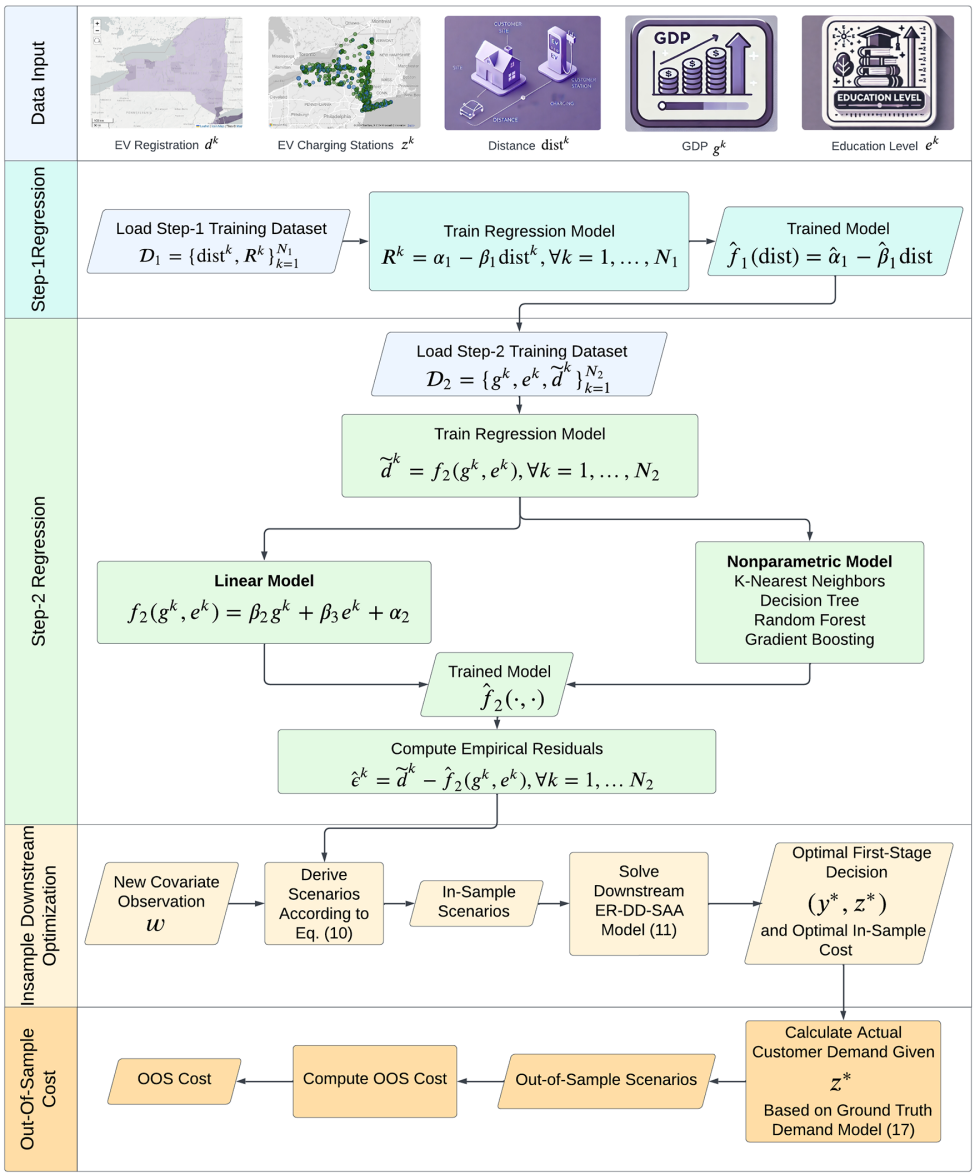


Figure 2: Flowchart of ER-DD-SAA framework for solving EV charging location problem.

Table 1: An illustrative example to show time-series data used in the Step-1 training dataset.

Day	$\boldsymbol{z}$	$\mathbb{E}[d_j(\boldsymbol{z}, \boldsymbol{w})]$
D_1	$\boldsymbol{z} = (0, 0, 0)$	$\bar{d}_j + f_2(\text{g}_j, \text{e}_j)$
D_2	$\boldsymbol{z} = (2, 0, 0)$	$\bar{d}_j + 2 \times \min\{D_2 - D_1, T\} \times e^{\alpha_1 - \beta_1 \text{ dist}_{1j}} + f_2(\text{g}_j, \text{e}_j)$
D_3	$\boldsymbol{z} = (2, 5, 0)$	$\bar{d}_j + 2 \times \min\{D_2 - D_1, T\} \times e^{\alpha_1 - \beta_1 \text{ dist}_{1j}} + 5 \times \min\{D_3 - D_2, T\} \times e^{\alpha_1 - \beta_1 \text{ dist}_{2j}} + f_2(\text{g}_j, \text{e}_j)$

$i \in I$ can be calculated as follows:

$$\Delta d_j(\boldsymbol{z}, \boldsymbol{w}) = \min\{t_i, T\} e^{\alpha_1 - \beta_1 \text{dist}_i} z_i, \tag{10}$$

where the left-hand side is the difference in EV registrations between two consecutive time stamps. Dividing $\min\{t_i, T\}z_i$ on both sides of (10) and taking logarithm, we have

$$\log\left(\frac{\Delta d_j(\boldsymbol{z}, \boldsymbol{w})}{\min\{t_i, T\}z_i}\right) = \alpha_1 - \beta_1 \text{dist}_{ij}, \tag{11}$$

which can be effectively regressed using linear regression methods. Note that we omit the zero-mean error term on the right-hand side of Eq. (10) and (11) that would be present in typical exponential regression. Denoting the response variable as $R_{ij} := \log\left(\frac{\Delta d_j(\boldsymbol{z},\boldsymbol{w})}{\min\{t_i,T\}z_i}\right)$ (i.e., the left-hand side of (11)), we group the data points $(\text{dist}_{ij}, R_{ij})$ for all $i \in I$ and $j \in J$ over a period of time (within one year) into our Step-1 training dataset $\mathcal{D}_1$ and renumber them as $\mathcal{D}_1 = \{\text{dist}^k, R^k\}_{k=1}^{N_1}$, where dist^k represents the distance between a charging station and a customer site for data point k, and R^k denotes the corresponding response for data point k, and N_1 is the number of data points used for Step-1 regression after the data preprocessing as described above. Unlike previous studies that assign fixed values to α_1 and β_1 (Basciftci et al. (2021); Mahmutoğulları and Yaman (2023)), we estimate these parameters using real-world data and linear regression techniques by solving the following optimization problem:

$$\{\hat{\alpha}_1, \hat{\beta}_1\} = \arg\min_{\alpha_1, \beta_1} \frac{1}{N_1} \sum_{k=1}^{N_1} \ell(\alpha_1 - \beta_1 \text{dist}^k, R^k), \tag{12}$$

where $\ell(\cdot, \cdot)$ denotes the loss function, and we apply the ordinary least squares (OLS) method in our experiments. We denote the prediction model for Step-1 regression as $\hat{f}_1(\text{dist}) = \hat{\alpha}_1 - \hat{\beta}_1 \text{dist}$.

3.3 Step-2 Regression

After obtaining the estimated parameters $\hat{\alpha}_1, \hat{\beta}_1$ from Step-1 regression, we substitute them into (9). We then consider both linear regression and nonparametric regression models for function $f_2(\cdot, \cdot)$. Since we can only obtain annual data for GDPs and education levels at a county level, in this step, we collect one data point for each county in each year and construct a Step-2 dataset $\mathcal{D}_2$ across multiple years, which has a different granularity than Step-1 dataset $\mathcal{D}_1$.

Step-2 Regression with Linear Model. We first consider the parametric linear regression model $f_2(\text{g}_j, \text{e}_j) = \alpha_2 + \beta_2 \text{g}_j + \beta_3 \text{e}_j$. Following Step-1 regression, Step-2 regression aims to estimate coefficients α_2, β_2 and β_3 of this model. Once we fix the parameters $\hat{\alpha}_1$, $\hat{\beta}_1$ obtained from Step-1 regression, Step-2 regression model can be expressed as

$$d_j(\boldsymbol{z}, \boldsymbol{w}) - \bar{d}_j - \sum_{i \in I} \min\{t_i, T\} e^{\hat{\alpha}_1 - \hat{\beta}_1 \text{dist}_{ij}} z_i = \alpha_2 + \beta_2 \text{g}_j + \beta_3 \text{e}_j + \epsilon, \quad \forall j \in J. \tag{13}$$

Denoting the left-hand side of (13) as response variable $\widetilde{d}_j$, we group the data points $(\text{g}_j, \text{e}_j, \widetilde{d}_j)$ among all customer sites $j \in J$ over multiple years (e.g., 4 years) into our Step-2 training dataset $\mathcal{D}_2$ and renumber them as $\mathcal{D}_2 = \{\text{g}^k, \text{e}^k, \widetilde{d}^k\}_{k=1}^{N_2}$. To estimate the parameters α_2, β_2 and β_3, we

solve the following minimization problem using OLS:

$$\{\hat{\alpha}_2, \hat{\beta}_3, \hat{\beta}_3\} = \underset{\alpha_2,\beta_2,\beta_3}{\arg\min} \frac{1}{N_2} \sum_{k=1}^{N_2} \ell(\alpha_2 + \beta_2 \mathrm{g}^k + \beta_3 \mathrm{e}^k, \widetilde{d}^k). \tag{14}$$

We denote this prediction model as $\hat{f}_2(\mathrm{g}, \mathrm{e}) = \hat{\alpha}_2 + \hat{\beta}_2 \mathrm{g} + \hat{\beta}_3 \mathrm{e}$.

However, the relationship between the response and the features may not be exactly linear. To address this, we consider nonparametric regression methods next.

Step-2 Regression with Nonparametric Model. To describe a more general and possibly nonlinear function $f_2(\cdot, \cdot)$, we adopt nonparametric regression models including k-Nearest Neighbors (kNN), decision tree, random forest, and gradient boosting. We introduce each of these methods below.

1. **kNN** (Altman, 1992) is a nonparametric method that predicts the outcome by averaging the response across k-closest training data points. A kNN estimator is given by

$$\hat{f}_2(\mathrm{g}, \mathrm{e}) = \frac{1}{K} \sum_{k \in \mathcal{N}_K(\mathrm{g},\mathrm{e})} \widetilde{d}^k,$$

   where K is the number of neighbors, $\mathcal{N}_K(\mathrm{g}, \mathrm{e}) := \Big\{ i = 1, \ldots, N_2 : \sum_{j=1}^{N_2} \mathbb{I}\big\{ \|(\mathrm{g}, \mathrm{e}) - (\mathrm{g}^i, \mathrm{e}^i)\| \geq \|(\mathrm{g}, \mathrm{e}) - (\mathrm{g}^j, \mathrm{e}^j)\| \big\} \leq K \Big\}$ is the K-nearest neighborhood of covariate information (g, e), $\mathbb{I}$ is the indicator function, and $\widetilde{d}^k$ is the corresponding response variable for data point k.

2. **Decision tree** (Breiman, 2017) is a tree-structured model that recursively partitions the feature space into sub-regions and assigns a prediction to a sub-region based on the average response in that region. A decision tree estimator is given by

$$\hat{f}_2(\mathrm{g}, \mathrm{e}) = \sum_{m=1}^{N_S} \mu_m \mathbb{I}\big\{ (\mathrm{g}, \mathrm{e}) \in S_m \big\},$$

   where the feature space is partitioned into N_S disjoint sub-regions: $\{S_1, S_2, \ldots, S_{N_S}\}$, $\mu_m = \frac{1}{|S_m|} \sum_{k=1}^{N_2} \widetilde{d}^k \mathbb{I}\{(\mathrm{g}^k, \mathrm{e}^k) \in S_m\}$ is the average response in sub-region S_m, and $|S_m|$ represents the cardinality of sub-region S_m.

3. **Random forest** (Breiman, 2001) is an ensemble learning method that averages predictions from multiple decision trees trained on bootstrap samples to reduce variance. Specifically, given a dataset $\mathcal{D}$, we generate N_T random subsets of the training data with replacement, called bootstrap samples. A random forest estimator is then given by

$$\hat{f}_2(\mathrm{g}, \mathrm{e}) = \frac{1}{N_T} \sum_{t=1}^{N_T} \hat{f}_2^{(t)}(\mathrm{g}, \mathrm{e}),$$

where N_T is the number of decision trees and $\hat{f}_2^{(t)}(\mathbf{g}, \mathbf{e})$ is the prediction from the decision tree trained on the t-th bootstrap subset.

4. **Gradient boosting** (Friedman, 2001) is an ensemble technique that sequentially builds weak learners to model the residuals of previous predictions. Unlike bagging methods such as random forest that reduce variance by averaging predictions from independent trees trained on bootstrap samples in parallel, gradient boosting takes an iterative approach to combining weak learners that individually do only slightly better than random chance. The algorithm starts with a simple initial model $\hat{f}_2^{(0)}(\mathrm{g}, \mathrm{e})$, often just the mean of the target variable. At each iteration m, the model is updated as

$$\hat{f}_2^{(m)}(\mathrm{g}, \mathrm{e}) = \hat{f}_2^{(m-1)}(\mathrm{g}, \mathrm{e}) + \gamma_m \psi_m(\mathrm{g}, \mathrm{e}),$$

where $\hat{f}_2^{(m-1)}(\mathrm{g}, \mathrm{e})$ is the model prediction from the previous iteration, γ_m is the learning rate of iteration m, and $\psi_m(\mathrm{g}, \mathrm{e})$ is a new simple model to predict the difference between the real value and predicted value from the previous step.

Due to the decision dependency modeled through an exponential function, we first perform linear regression on endogenous factors in Step 1. Then, in Step 2, we apply various nonparametric regression methods to study the impact of exogenous factors. This approach allows us to leverage the flexibility of nonparametric models to better capture the underlying relationships in the data during Step 2, while maintaining the interpretability provided by the parametric model in Step 1. Note that other nonparametric regression models can also be applied in Step 2 without altering the methodology.

3.4 ER-DD-SAA to EV Charging Station Location Problem

To account for the estimation error in the predicted regression models, we adopt ER-DD-SAA framework to incorporate empirical residuals—which quantify the difference between observed data points and predicted values—into our downstream optimization problem.

Specifically, from Step-2 regression, we calculate empirical residuals as follows:

$$\hat{\epsilon}^k \leftarrow \widetilde{d}^k - \hat{f}_2(\mathrm{g}^k, \mathrm{e}^k), \ \forall k = 1, \ldots, N_2. \tag{15}$$

Each empirical residual $\hat{\epsilon}^k$ is then added to the point estimator (i.e., predicted demand from the regression model) to construct a scenario, denoted as $\hat{d}_j(\boldsymbol{z}, \boldsymbol{w}, \hat{\epsilon}^k)$ for all $k = 1, \ldots, N_2, \ j \in J$. Given any first-stage decision $\boldsymbol{z}$ and covariates $\boldsymbol{w}$, the EV demand at customer site $j \in J$ under the k-th scenario is formulated as follows:

$$\hat{d}_j(\boldsymbol{z}, \boldsymbol{w}, \hat{\epsilon}^k) = \bar{d}_j + \sum_{i \in I} T e^{\hat{\alpha}_1 - \hat{\beta}_1 \mathrm{dist}_{ij}} z_i + \hat{f}_2(\mathrm{g}_j, \mathrm{e}_j) + \hat{\epsilon}^k, \ \forall k = 1, \ldots, N_2, \ j \in J. \tag{16}$$

Note that in (16), we assume that all the time intervals between two consecutively opened charging

stations t_i are greater than or equal to the threshold T (i.e., $\min\{t_i, T\} = T$) to simplify the downstream optimization model. This can represent a practical situation of deploying EV charging stations over multiple years while ensuring adequate intervals between each installation. One can also relax this assumption and optimize the timing of building EV charging stations in the first stage, which complicates the optimization model. For simplicity and practical deployment, we continue with this assumption.

Given these N_2 scenarios $\{\hat{\boldsymbol{d}}(\boldsymbol{z}, \boldsymbol{w}, \hat{\epsilon}^k)\}_{k=1}^{N_2}$, we construct the following ER-DD-SAA to approximate our problem (1)

$$\min_{\boldsymbol{y},\boldsymbol{z}} \quad \sum_{i\in I}\Big(f_i y_i + b_i z_i\Big) + \frac{1}{N_2}\sum_{k=1}^{N_2} h\big(\boldsymbol{z}, \hat{\boldsymbol{d}}(\boldsymbol{z}, \boldsymbol{w}, \hat{\epsilon}^k)\big) \tag{17a}$$

$$\text{s.t.} \quad z_i \le M_i y_i, \quad \forall i \in I, \tag{17b}$$

$$y_i \in \{0,1\},\ z_i \in \mathbb{Z}, \quad \forall i \in I, \tag{17c}$$

where we replace the expected second-stage cost $\mathbb{E}\big[h(\boldsymbol{z}, \boldsymbol{d}(\boldsymbol{z}, \boldsymbol{w}))|\boldsymbol{Z} = \boldsymbol{z}, \boldsymbol{W} = \boldsymbol{w}\big]$ with sample average approximation over N_2 scenarios, and the second-stage cost under each scenario k is given by

$$h\big(\boldsymbol{z}, \hat{\boldsymbol{d}}(\boldsymbol{z}, \boldsymbol{w}, \hat{\epsilon}^k)\big) = \min_{\boldsymbol{x},\boldsymbol{s}} \quad \sum_{i\in I}\sum_{j\in J} c_{ijk}x_{ijk} + \sum_{j\in J}\Big(p_j s_{jk} - r_j \hat{d}_j(\boldsymbol{z}, \boldsymbol{w}, \hat{\epsilon}^k)\Big) \tag{18a}$$

$$\text{s.t.} \quad \sum_{i\in I} x_{ijk} + s_{jk} = \hat{d}_j(\boldsymbol{z}, \boldsymbol{w}, \hat{\epsilon}^k), \quad \forall j \in J, \tag{18b}$$

$$\sum_{j\in J} x_{ijk} \le C_i z_i, \quad \forall i \in I, \tag{18c}$$

$$x_{ijk}, s_{jk} \ge 0, \quad \forall i \in I,\ \forall j \in J. \tag{18d}$$

Here, x_{ijk} and s_{jk} represent the amount of demand at customer site $j \in J$ fulfilled by station $i \in I$ and unsatisfied demand at site $j \in J$ under scenario k, respectively.

4 Numerical Results Based on Synthetic Datasets

In this section, we run numerical experiments based on synthetic datasets to (i) first test our proposed regression procedure under a controlled environment, and to (ii) better conduct out-of-sample tests and sensitivity analyses. We report the regression results in Section 4.1 and compare the performance of our proposed two-step regression approach with the one-step joint regression in Section 4.2. We then discuss the downstream optimization results in Section 4.3. Finally, we vary the problem parameters of the downstream optimization model and the decision-dependency parameters of the ground truth demand model in order to conduct sensitivity analyses and to gain further insights into the behavior of the EV charging station location problem in Section 4.4.

We perform all regression analyses using scikit-learn 1.2.1 and solve all optimization models

using Gurobi 10.0.2 with a 1% MIP gap, both implemented in Python 3.10.12. All tests are conducted on a system equipped with an Apple M1 Pro Chip.

4.1 Regression Results

We generate synthetic datasets based on a ground truth decision-dependent demand model in Section 4.1.1. We introduce a decision-independent regression model as a benchmark and compare its performance with our two-step regression approach in Section 4.1.2, respectively.

4.1.1 Synthetic Datasets

To generate synthetic datasets, we first assume a ground truth demand model as follows

$$d_j(\boldsymbol{z}, \boldsymbol{w}) = \bar{d}_j + \sum_{i \in I} e^{\alpha_1 - \beta_1 \text{dist}_{ij}} z_i + \alpha_2 + \beta_2 \text{g}_j + \beta_3 \text{e}_j + \epsilon, \ \forall j \in J, \tag{19}$$

where $\alpha_1 = 4$, $\beta_1 = 0.05$, $\alpha_2 = 1500$, $\beta_2 = 3000$, $\beta_3 = 300$, and $\epsilon \sim \mathcal{N}(0, 10^2)$ in the default setting. Note that we set $T = 1$ and $f_2(\text{g}_j, \text{e}_j)$ to be a linear function in (9) to simplify the regression and downstream optimization models. We will add these complexities back later in the case study section. Initial demand $\bar{d}_j$ is uniformly sampled from the range $[10000, 180000]$ for each customer site $j \in J$. Here, the magnitude of α_1 and β_1 controls the degree of decision dependency in the ground truth demand model. We will vary these parameters in the regression and downstream optimization analyses to illustrate the value of endogenous factors under different degrees of decision dependency. The synthetic dataset for the regression analysis is then generated based on Eq. (19), where we consider 300 EV charging stations with randomly assigned different opening dates within 6 years and 50 customer sites over a geographical area centered at the coordinate $(40, -73)$, within a 45-mile radius (see Figure 5). We assume that both GDPs and education levels increase over time. To generate these exogenous factors for each customer site $j \in J$, we first assign baseline GDP and education level values by uniformly sampling from the ranges $[6000, 14000]$ and $[4\%, 16\%]$, respectively. We assume that GDP increases by $16,000$ per year and education level increases by 4% per year. For each customer site $j \in J$ and each year, the GDP and education level are computed as the sum of their respective baseline values, the incremental changes, and error terms $\epsilon^g \sim \mathcal{N}(0, 12000^2)$ and $\epsilon^e \sim \mathcal{N}(0, 0.02^2)$, respectively. The final GDP and education level values for each customer site and year are then normalized to the $[0, 1]$ range. For each charging station, we compute the corresponding demand $d_j(\boldsymbol{z}, \boldsymbol{w})$ at each customer site j on the opening date following Eq. (19), creating a dataset $\{\text{dist}^k, \text{g}^k, \text{e}^k, \boldsymbol{z}^k, \bar{d}^k, d^k\}_{k=1}^N$ with $N = 5831$. This dataset is divided into a training set comprising 3865 data points from the first four years and a testing set containing 1966 data points from the last two years.

4.1.2 Regression Results Analyses

We compare our decision-dependent regression model (19) with a decision-independent counterpart, where customer demand is predicted solely based on exogenous factors without considering the

impact of location/capacity decisions. Therefore, the customer demand at customer site $j \in J$ in the decision-independent regression benchmark is given by

$$d_j^{DI}(\mathbf{g}, \mathbf{e}) = \bar{d}_j + \alpha_2^{DI} + \beta_2^{DI} \mathrm{g}_j + \beta_3^{DI} \mathrm{e}_j + \epsilon^{DI}, \ \forall j \in J, \tag{20}$$

where we use superscript DI to denote the relevant terms in this model. Next, we compare the regression results between these two models.

Step-1 Regression Results. Following the proposed two-step regression approach outlined in Section 2, we first implement the Step-1 regression on the synthetic dataset. The response variable R_{ij} is calculated by measuring the incremental change in customer demand from two consecutively opened stations, divided by the number of chargers, i.e., $R_{ij} := \log\left(\Delta d_j(\boldsymbol{z}, \boldsymbol{w})/z_i\right)$. We remove data points with $\Delta d_j(\boldsymbol{z}, \boldsymbol{w}) \leq 0$. Grouping such data points among all customer sites and charging stations over 4 years yields our Step-1 training dataset $\mathcal{D}_1 = \{\mathrm{dist}^k, R^k\}_{k=1}^{N_1}$ with size N_1. Using OLS, the Step-1 regression results are shown in Table 2, where we vary the parameter α_1 in the ground truth demand model (19) to reflect different degrees of decision dependency and mark the best results in bold. From Table 2, as α_1 increases, the ground truth demand model becomes more decision-dependent, and the performance of the Step-1 regression improves progressively with lower mean squared error (MSE), mean absolute error (MAE) and maximum error (ME) and a higher R^2. Figure 3 illustrates the results of our Step-1 regression with the default setting ($\alpha_1 = 4$), where Figure 3a reveals a clearly decreasing trend of the response variable with respect to the distance.

Table 2: Step-1 regression results for synthetic experiments.

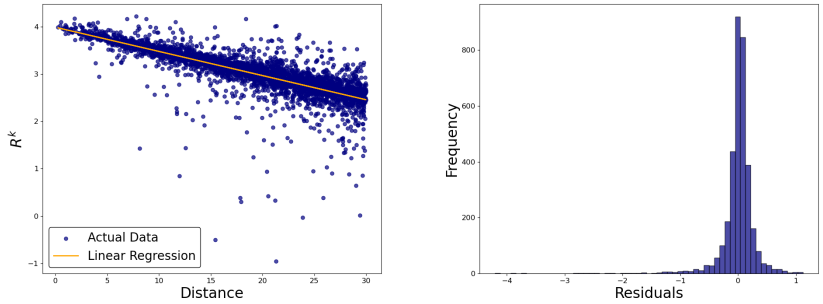
α_1	Data Size N_1	Intercept $\hat{\alpha}_1$	Coefficient $\hat{\beta}_1$	MSE	MAE	ME	R^2
-1	1808	0.5342	0.0011	1.5941	0.7418	7.3624	0.0000
0	1962	0.6131	0.0050	1.6942	0.7368	8.5583	0.0008
1	2280	0.9406	0.0129	1.3390	0.6295	7.6032	0.0069
2	2854	1.8469	0.0376	0.8835	0.4653	9.4805	0.0807
3	3299	2.9833	0.0515	0.3180	0.2335	6.0064	0.3094
4	3446	3.9913	0.0511	0.0946	0.0910	4.2039	0.5968
5	**3470**	**5.0007**	**0.0504**	**0.0116**	**0.0329**	**1.4562**	**0.9212**

Step-2 Regression Results. After obtaining estimated parameters $\hat{\alpha}_1$ and $\hat{\beta}_1$, we construct the response variable $\widetilde{d}_j := d_j(\boldsymbol{z}, \boldsymbol{w}) - \bar{d}_j - \sum_{i \in I} e^{\hat{\alpha}_1 - \hat{\beta}_1 \mathrm{dist}_{ij}} z_i$ and regress the following linear model

$$\widetilde{d}_j := d_j(\boldsymbol{z}, \boldsymbol{w}) - \bar{d}_j - \sum_{i \in I} e^{\hat{\alpha}_1 - \hat{\beta}_1 \mathrm{dist}_{ij}} z_i = \alpha_2 + \beta_2 \mathrm{g}_j + \beta_3 \mathrm{e}_j + \epsilon, \ \forall j \in J, \tag{21}$$

where $d_j(\boldsymbol{z}; \boldsymbol{w})$ comes from the training dataset calculated based on the ground truth model (19), I contains all the charging stations opened before the corresponding time for each data point, and the term $\sum_{i \in I} e^{\hat{\alpha}_1 - \hat{\beta}_1 \mathrm{dist}_{ij}} z_i$ measures the total demand contribution by endogenous factors.

Grouping the data points over all customer sites and time stamps yields a Step-2 training dataset $\mathcal{D}_2 = \{\mathrm{g}^k, \mathrm{e}^k, \widetilde{d}^k\}_{k=1}^{N_2}$ with $N_2 = 3865$. We perform linear regression on $\mathcal{D}_2$ to get the estimated



(a) Response $\boldsymbol{R}$ versus feature `dist`. (b) Residuals frequency plot.

Figure 3: Step-1 regression analysis for synthetic experiments in the default setting: graph (a) shows the response $\boldsymbol{R}$ as a function of feature `dist`, while graph (b) illustrates the residual distribution to evaluate model performance.

parameters $\hat{\alpha}_2, \hat{\beta}_2, \hat{\beta}_3$. Then, the empirical residuals can be computed as

$$\hat{\epsilon}^k := \widetilde{d}^k - (\hat{\alpha}_2 + \hat{\beta}_2 \mathrm{g}^k + \hat{\beta}_3 \mathrm{e}^k), \ \forall k = 1, \ldots, N_2. \tag{22}$$

On the other hand, for the decision-independent regression model (20), since it does not involve the decision $\boldsymbol{z}$, we do not need to perform Step-1 regression and can instead construct a training dataset $\mathcal{D}_2 = \{\mathrm{g}^k, \mathrm{e}^k, \widetilde{d}^{k,DI}\}_{k=1}^{N_2}$ where $\widetilde{d}^{k,DI} = d^k - \bar{d}^k$. We first estimate the coefficients in this model as $\hat{\alpha}_2^{DI}, \hat{\beta}_2^{DI}, \hat{\beta}_3^{DI}$ and then compute the empirical residuals as

$$\hat{\epsilon}^{k,DI} := \widetilde{d}^{k,DI} - (\hat{\alpha}_2^{DI} + \hat{\beta}_2^{DI} \mathrm{g}^k + \hat{\beta}_3^{DI} \mathrm{e}^k)), \ \forall k = 1, \ldots, N_2. \tag{23}$$

Table 3 compares the predictive performance of the decision-dependent and decision-independent regression models across varying levels of α_1. As before, the better values of the evaluation metrics, along with their corresponding estimated coefficients, are highlighted in bold. At lower values of α_1 (e.g., $\alpha_1 = -1, 0$), the decision-independent model achieves lower MSE, MAE, and ME and a higher R^2 in the testing datasets, because there is less decision dependency in the ground truth demand model. However, as α_1 increases (when $\alpha_1 \geq 1$), the decision-dependent model outperforms the decision-independent counterpart, obtaining much smaller MSE, MAE, and ME and much higher R^2, as well as more accurate estimates for coefficients $\hat{\alpha}_2$, $\hat{\beta}_2$, $\hat{\beta}_3$. The results further indicate that, the decision-dependent model in low-decision-dependency setting performs relatively better compared to decision-independent model in high-decision-dependency setting. To see this, examine in contrast the first and last rows of Table 3 (i.e., R^2 is 0.58 for the decision-dependent model in low-decision-dependency setting versus 0.06 for the decision-independent model in high-decision-dependency setting).

In summary, the regression results in Table 3 demonstrate that, under a ground truth demand model with moderate to high decision dependency, our decision-dependent regression model (9) outperforms the decision-independent benchmark by achieving much lower error and thus should

Table 3: Step-2 regression results for synthetic experiments.

α_1	Decision Dependency	$\hat{\alpha}_2$	$\hat{\beta}_2$	$\hat{\beta}_3$	MSE (Test)	MAE (Test)	ME (Test)	R^2 (Test)
-1	Decision-Dependent	1,566.35	2,454.68	-224.76	47,779.36	160.80	583.19	0.58
	Decision-Independent	**1,492.91**	**3,059.09**	**351.07**	**564.82**	**17.37**	**73.38**	**0.99**
0	Decision-Dependent	1,555.25	2,548.78	-140.08	33,695.66	132.4419	483.24	0.68
	Decision-Independent	**1,481.18**	**3,161.05**	**437.61**	**3,565.10**	**46.42**	**172.78**	**0.97**
1	Decision-Dependent	**1,538.37**	**2,695.18**	**-14.48**	**17,600.71**	**88.91**	**337.70**	**0.82**
	Decision-Independent	1,449.27	3,438.20	672.84	25,809.59	126.25	459.17	0.84
2	Decision-Dependent	**1,506.65**	**2,956.75**	**235.20**	**1,298.07**	**21.57**	**108.66**	**0.98**
	Decision-Independent	1,362.53	4,191.58	1,312.27	190,379.90	346.44	1,243.89	0.52
3	Decision-Dependent	**1,485.07**	**3,122.17**	**412.84**	**2,306.65**	**35.14**	**134.58**	**0.98**
	Decision-Independent	1,126.76	6,239.47	3,050.43	1,406,955.13	943.07	3,376.99	0.22
4	Decision-Dependent	**1,473.69**	**3,215.84**	**500.93**	**7,135.06**	**61.83**	**229.11**	**0.94**
	Decision-Independent	485.87	11,806.23	7,775.24	10,397,809.06	2,567.73	9,175.34	0.10
5	Decision-Dependent	**1,485.52**	**3,113.66**	**413.16**	**2,330.25**	**32.74**	**126.42**	**0.98**
	Decision-Independent	-1,256.24	26,938.24	20,618.58	76,835,841.17	6,968.85	24,936.91	0.06

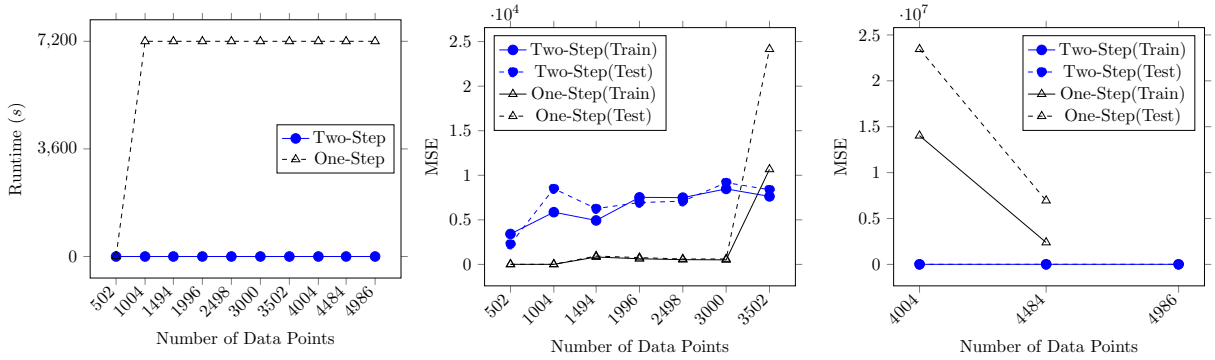
be preferred under such settings. Moreover, our proposed two-step regression approach closely estimates the coefficients in the ground truth model in these synthetic experiments, indicating the effectiveness of our proposed approximate approach.

4.2 Comparison between Two-Step Regression and One-Step Joint Regression

In this section, we compare the performance of our proposed two-step regression approach with the one-step regression approach by minimizing the MSE jointly. To better observe the trends when we increase the number of data points, we decrease the problem size and consider 30 EV charging stations and 30 customer sites over a geographical area centered at the coordinate (40, -73) within a 45-mile radius. We generate multiple sequences of opening dates for the charging stations and combine data from these different sequences to construct larger datasets. Specifically, for the one-step joint regression, we aim to minimize the overall MSE for the dataset $\{(\text{dist}^n, \text{g}^n, \text{e}^n, z^n, \bar{d}^n, d^n)\}_{n=1}^N$ as follows

$$\min_{\alpha_1,\beta_1,\alpha_2,\beta_2,\beta_3} \frac{1}{N}\sum_{n=1}^{N}\left(d^n - \left(\bar{d}^n + \sum_{i\in I_n} e^{\alpha_1-\beta_1\text{dist}^n} z_i + \alpha_2 + \beta_2\text{g}^n + \beta_3\text{e}^n\right)\right)^2, \tag{24}$$

where I_n represents the set of existing EV charging stations that influence the customer site associated with data point n. Model (24) is highly nonconvex and we solve it using the Nonconvex solver in Gurobi 12.0.1 with the time limit set to 2 hours. After the time limit, we record the best solution found by Gurobi and report its corresponding MSE. We display the runtime and MSE for the one-step regression approach and the proposed two-step regression approach in Figure 4 when we increase the number of data points from 502 to 4986. The detailed performance of parameter estimation is illustrated in Figure 11 in Appendix C.



(a) Runtime (b) MSE for Smaller Datasets (c) MSE for Larger Datasets

Figure 4: Performance comparisons between the one-step and two-step regression approach (Note that the scales for the y-axis between Plots (b) and (c) are different).

We observe that the one-step regression approach reaches the time limit of 2 hours when solving datasets larger than 502 data points. Furthermore, its MSE becomes excessively large for datasets exceeding 3502 data points, and it fails to obtain a feasible solution for the largest dataset (4986 data points) within the time limit. In contrast, the proposed two-step regression approach solves all problem instances within one second. Even though it performs worse than the one-step approach in smaller datasets, it achieves a much lower MSE in larger datasets, demonstrating its efficiency and effectiveness in real-world applications. Moreover, unlike the one-step regression approach, our proposed two-step method can incorporate non-parametric regression models in decision-dependent settings, as illustrated in Section 5.1.2. In the sequel, we will utilize the regression results from the two-step approach in Section 4.1 and report its corresponding downstream optimization results.

4.3 Downstream Optimization Results

In the default setting, we consider $I = 50$ potential EV charging station locations with a maximum number of chargers $M_i = 10$ and $J = 5$ customer sites, each within a radius of 15 miles (see Figure 5). The fixed cost for opening an EV charging station f_i is set at 8,000 for all $i \in I$. The unit installation fee b_i to install a charger at any location $i \in I$ is 400. The unit transportation cost c_{ij} is defined as 0.005 times the distance between the EV charging station $i \in I$ and customer site $j \in J$. For each customer site $j \in J$, the unit revenue r_j is set to 10, and the unit penalty for unmet demand p_j is set to 5. Each charger at location $i \in I$ has a capacity $C_i = 500$. We will also vary these parameters and conduct sensitivity analyses in Section 4.4. Because of the large training dataset ($I = 50$ and $N_2 = 3865$), for computational efficiency, we sample 1000 scenarios from the Step-2 residual set $\{\hat{\epsilon}^k\}_{k=1}^{3865}$ and use them to construct in-sample scenarios $\hat{d}_j(\boldsymbol{z}, \boldsymbol{w}, \hat{\epsilon}^k) = \bar{d}_j + \sum_{i \in I} e^{\hat{\alpha}_1 - \hat{\beta}_1 \text{dist}_{ij}} z_i + \hat{\alpha}_2 + \hat{\beta}_2 \text{g}_j + \hat{\beta}_3 \text{e}_j + \hat{\epsilon}^k$. These scenarios are plugged into (18) to form the ER-DD-SAA model to obtain optimal first-stage decisions.

Benchmark: ER-DI-SAA. By incorporating the residuals $\{\hat{\epsilon}^{k,DI}\}_{k=1}^{N_2}$ obtained from the decision-independent regression model into the point estimator, we construct decision-independent in-sample

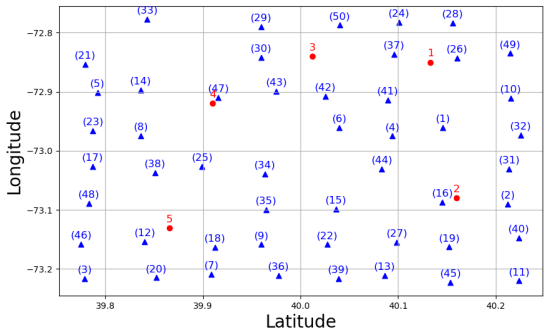


Figure 5: Locations of customer sites and potential EV charging stations used in synthetic experiments. Red dots represent customer sites, while blue triangles indicate potential EV charging stations.

scenarios $\hat{d}_j^{DI}(\mathbf{g}, \mathbf{e}, \hat{\epsilon}^{k,DI}) = \bar{d}_j + \hat{\alpha}_2^{DI} + \hat{\beta}_2^{DI} \mathrm{g}_j + \hat{\beta}_3^{DI} \mathrm{e}_j + \hat{\epsilon}^{k,DI}$. Plugging these scenarios into Model (18) gives the ER-DI-SAA optimization model. Note that this model has the same number of decision variables and constraints as the ER-DD-SAA model where the only difference is that the demand scenarios $\hat{d}_j^{DI}(\mathbf{g}, \mathbf{e}, \hat{\epsilon}^{k,DI})$ are independent of the first-stage decision $\boldsymbol{z}$.

Since our synthetic experiment fully specifies the actual customer demand $d_j(\boldsymbol{z}, \boldsymbol{w})$ based on the ground truth demand model (19), we construct 5000 out-of-sample demand scenarios given any first-stage location and capacity decisions and calculate the average out-of-sample (OOS) cost across 10 simulation runs. Table 4 compares the in-sample and out-of-sample performance between ER-DD-SAA and ER-DI-SAA.

Table 4: Optimal first-stage solutions, in-sample cost, and OOS cost for synthetic experiments.

Optimization	Location and Capacity Decision	# Chargers	In-Sample Cost	OOS Cost
ER-DD-SAA	[1, 2, 4, 6, 7, 8, 9, 10, 12, 14, 15, 16, 18, 19, 22, 24, 25, 26, 27, 28, 29, 30, 31, 32, 34, 35, 36, 37, 38, 41, 42, 43, 44, 47, 49, 50]: 10 chargers	**360**	**-956,351.89**	**-963,289.34**
ER-DI-SAA	[2, 7, 8, 12, 14, 16, 18, 19, 20, 24, 25, 26, 28, 29, 30, 31, 37, 38, 40, 41, 42, 43, 47, 50]: 10 chargers	240	-737,918.00	-919,490.32

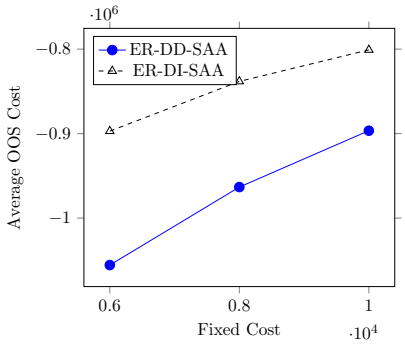
Table 4 reveals that the ER-DD-SAA model recommends installing a higher number of chargers in the first stage compared to ER-DI-SAA, yielding a higher demand and better in-sample and out-of-sample performance. While ER-DI-SAA chooses to build EV charging stations in locations #20 and #40, ER-DD-SAA picks locations that are closer to most customer sites, such as #1, 4, 6, 9, 10, 15, which can bring more demand according to the distance effect in Eq. (9).

4.4 Sensitivity Analyses

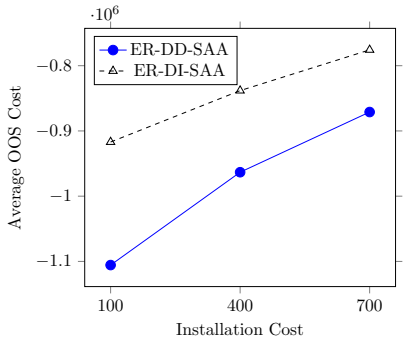
We now investigate the impact of various problem parameters on the performance of the EV charging station location optimization model. Specifically, we focus on the effect of unit fixed cost,

unit charger installment cost, unit revenue per customer, transportation cost per mile, unit penalty for unmet demand, and capacity per charger. Figure 6 shows the average OOS cost of our ER-DD-SAA model and its decision-independent counterpart (ER-DI-SAA) across 10 simulation runs. In all graphs, the default values appear in the middle.

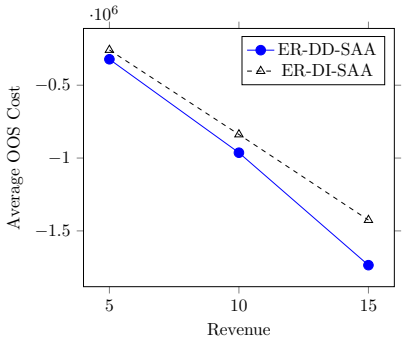
Figures 6a and 6b show that both models are sensitive to the variability of fixed cost and installation cost. As these costs increase, the performance of both models worsens. Figure 6c demonstrates that both models are sensitive to unit revenue, with the performance gap between ER-DD-SAA and ER-DI-SAA widening in favor of ER-DD-SAA as the unit revenue increases. Figures 6d and 6e, on the other hand, reveal that the ER-DD-SAA model is less sensitive to transportation and penalty costs, while the ER-DI-SAA model is more sensitive to changes in penalty costs. Finally, Figure 6f illustrates that both models yield the same optimal first-stage decisions with a low capacity per charger (i.e., $C_i = 100$) because both models choose to build no chargers at all locations due to the low capacity and penalty. However, as capacity increases, both models perform better and ER-DD-SAA outperforms ER-DI-SAA by installing more EV chargers, which leads to higher customer demand.



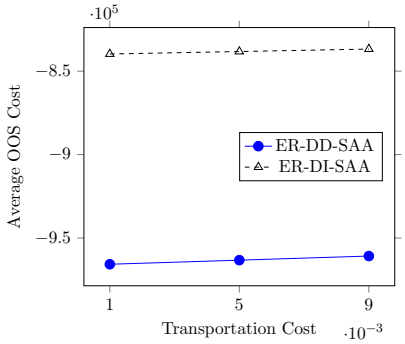
(a) Effect of fixed cost f_i on average OOS cost



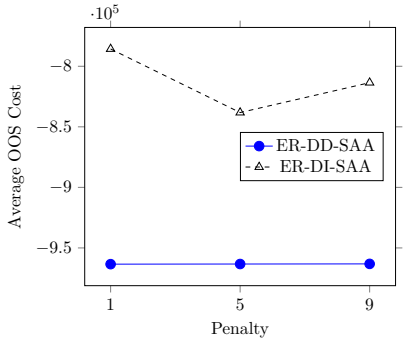
(b) Effect of charger installation fee b_i on average OOS cost



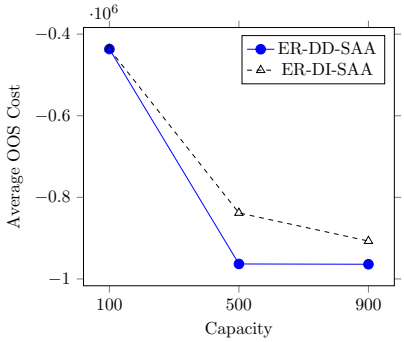
(c) Effect of revenue r_j on average OOS cost



(d) Effect of unit transportation cost c_{ij} on average OOS cost



(e) Effect of penalty p_j on average OOS cost

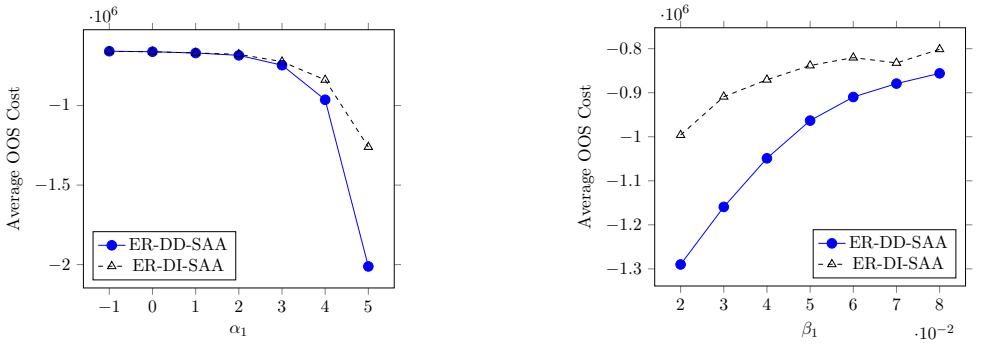


(f) Effect of capacity per charger C_i on average OOS cost

Figure 6: Effects of different parameters on the average OOS cost in the EV charging station location problem.

Next, we examine the impact of the degree of decision dependency on OOS costs, as illustrated

in Figure 7. Figure 7a demonstrates that at lower values of α_1, the performance of ER-DD-SAA and ER-DI-SAA closely matches each other. However, as α_1 increases, the ground truth model becomes more decision-dependent, and the gap between the two models becomes larger, with ER-DD-SAA progressively outperforming ER-DI-SAA. From Figure 7b, as β_1 increases, the gap between the two models narrows because the distance effect (i.e., $e^{\alpha_1 - \beta_1 \text{dist}_{ij}}$) becomes smaller in the ground truth demand model (19).



(a) Effect of α_1 on average OOS cost

(b) Effect of β_1 on average OOS cost

Figure 7: Effects of the degree of decision dependency on the average OOS cost in the EV charging station location problem.

5 A Case Study Based on Real-World Data in New York State

This section presents the computational results of our case study, based on real-world data in New York State from 2017 to 2022. Given the inherent complexity of real-world data and the unknown ground-truth relationship, the one-step regression approach fails to find a feasible solution within the time limit (i.e., 2 hours). Therefore, we focus exclusively on the two-step regression approach in this case study. In our analyses, we use the four years 2017–2020 (i.e., two-thirds of the data) as the training dataset and the remaining two years 2021–2022 (i.e., one-third of the data) as the testing dataset. We begin by analyzing the regression results of our decision-dependent regression model (9) and a decision-independent model in Section 5.1. Based on the regression results, we solve the downstream ER-DD-SAA model and compare the results against its decision-independent counterpart (ER-DI-SAA) in Section 5.2, to illustrate the value of endogenous factors.

5.1 Regression Results

In this section, we first outline the data sources and preprocessing in Section 5.1.1. Then, we report Step-1 and Step-2 regression results for both models in Section 5.1.2 to compare their predictive performance.

5.1.1 Data Sources and Preprocessing

We concentrate our case study on New York State, which has county-level data on EV registrations, GDPs, and education levels for all 62 counties from publicly available data sources. We consider both battery electric vehicles (BEV) and plug-in hybrid electric vehicles (PHEV). Between 2017 and 2022, there were 138,816 original EV registrations and 7,636 new charging ports installed in New York State (Atlas Public Policy). We use monthly EV registration data in each county as an indicator of EV demand at each customer site, acquiring county-level data for New York State over 2017–2022 from Atlas Public Policy (see a snapshot of EV registration data in Table 10). Additionally, we obtain data on EV charging stations from Center (2023), focusing on publicly accessible stations (see a snapshot of EV charging station data in Table 11). This dataset includes detailed information on each station's precise location (latitude and longitude) and opening date, making it essential for our Step-1 regression analysis. We collect county-level annual GDP data encompassing all industries from the Federal Reserve Economic Data (FRED) database at Federal Reserve Bank of St. Louis (b), which ranges from \$233,351 to \$885,651,810. We also acquire education data, specifically the percentage of the population with a bachelor's degree or higher, from the FRED database at Federal Reserve Bank of St. Louis (a), which ranges from 15.4% to 62.8%. Both datasets cover the same year range from 2017 to 2022.

In our experiments, each county serves as a customer site, with monthly EV registration data converted to daily estimates by dividing the monthly totals by the number of days in each month. To establish spatial relationships, we compute distances from each county to each EV charging station using the haversine distance metric. For each county, we construct time-series data (as illustrated in Table 1) to incorporate the opening dates of charging stations within a certain distance threshold. Due to the exponential decay effect with respect to distance in our demand model (1) (i.e., $\exp(\alpha_1 - \beta_1 \text{dist}_{ij})$), the demand increase contributed by the opening of a charging station outside of a certain radius becomes negligible. This exponential decay property implies that only EV charging stations opened within a specific radius around each customer site will meaningfully influence the customer demand there. Accordingly, we tuned this radius threshold in our experiments and found that a 30-mile threshold achieved the best performance. As a result, we only consider charging stations within 30 miles when estimating demand at each customer site. To keep a lower diagonal $\boldsymbol{z}$-matrix as in Table 1, we exclude data points where multiple charging stations were opened in different locations on the same day.

5.1.2 Regression Result Analyses

For the decision-dependent demand model, we adopt the general nonlinear regression model in Eq. (9). For the decision-independent counterpart, we drop the second term that depends on $\boldsymbol{z}$ and consider

$$d_j^{DI}(\mathbf{g}, \mathbf{e}) = \bar{d}_j + f_2^{DI}(\mathrm{g}_j, \mathrm{e}_j) + \epsilon^{DI}. \tag{25}$$

We implement both parametric and nonparametric regression methods to estimate the functions $f_2(\mathrm{g}_j, \mathrm{e}_j)$ and $f_2^{DI}(\mathrm{g}_j, \mathrm{e}_j)$ in decision-dependent and decision-independent demand models, respectively. We apply data from New York State to estimate the nonlinear regression model (9) following the two-step regression approach developed in Section 3. The regression results for Step 1 and Step 2 are presented below.

Step-1 Regression Results. Following Eq. (11), the response variable R_{ij} for Step-1 regression is calculated by measuring the incremental change in customer demand from two consecutively opened stations in the same county, divided by the number of chargers times $\min\{t_i, T\}$, i.e., $R_{ij} = \log\left(\frac{\Delta d_j(\boldsymbol{z},\boldsymbol{w})}{\min\{t_i,T\}z_i}\right)$. Grouping all such data points among all customer sites and charging stations from 2017 to 2020 yields our Step-1 training dataset $\mathcal{D}_1 = \{\mathrm{dist}^k, R^k\}_{k=1}^{N_1}$ with $N_1 = 1191$.

After removing outliers with z-scores[1] exceeding a threshold of 3, we solve optimization problem (12) using OLS. We tune the time threshold T from the set $\{1, 15, 30, 45, 60\}$. Table 5 summarizes the Step-1 regression results for different values of T, including the estimated coefficients ($\hat{\alpha}_1$, $\hat{\beta}_1$), mean squared error (MSE), median absolute error (MAE), max error (ME), and R-squared (R^2), where we mark the best results in bold.

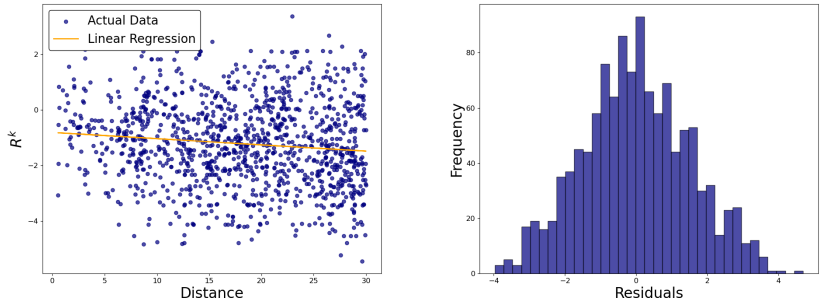
Table 5: Step-1 regression results for different time thresholds T.

T	Intercept $\hat{\alpha}_1$	Coefficient $\hat{\beta}_1$	MSE	MAE	ME	R^2
1	1.0210	0.0076	3.2010	1.1989	5.0503	0.0011
15	-0.6792	0.0194	2.3025	1.0558	4.2499	0.0097
30	-0.8132	0.0223	**2.2888**	1.0141	4.6860	0.0129
45	-0.8492	0.0237	2.3069	**1.0109**	**3.3482**	0.0144
60	-0.8622	0.0245	2.3312	1.0245	4.3316	**0.0151**

From Table 5, the time threshold $T = 30$ achieves the lowest MSE, while $T = 45$ achieves the lowest MAE and ME. All of the time thresholds yield a positive $\hat{\beta}_1$, indicating a decreasing trend of the demand contribution with respect to distance. Based on these results, we fix $T = 30$ with estimated coefficients $\{\hat{\alpha}_1, \hat{\beta}_1\} = \{-0.8132, 0.0223\}$ in the following analyses, because it provides the best MSE with comparatively good MAE, ME, and R^2 values.

Figure 8 illustrates the results of the Step-1 regression analysis with $T = 30$. Figure 8a depicts the response variable R^k (i.e., the logarithm of the demand increase adjusted by time duration and number of chargers) on the y-axis versus the distance between charging stations and customer sites on the x-axis. The blue dots represent the training data points, while the orange line indicates the fitted linear regression model. The observed negative trend suggests that as the distance between the charging station and the customer site increases, its impact on the demand decreases, highlighting the influence of proximity on customer demand. Figure 8b displays the histogram of the residuals from the regression model. The residuals are approximately centered around zero.

[1] A z-score measures how many standard deviations a data point is from the mean. It is calculated as $z = (x-\mu)/\sigma$, where x is the data point, μ is the mean, and σ is the standard deviation. Data points with absolute z-scores exceeding 3 are considered outliers and removed, retaining approximately 99.7% of normally distributed data.



(a) Response $\boldsymbol{R}$ versus feature `dist`. (b) Residuals frequency plot.

Figure 8: Step-1 regression analysis: graph (a) shows the response $\boldsymbol{R}$ as a function of feature `dist`, while graph (b) illustrates the residual distribution to evaluate model performance.

However, as can be seen from Table 5, for all time thresholds T, the R^2 scores are pretty low, suggesting that solely considering the endogenous factors is not sufficient to accurately predict the demand. Next, we fix the parameters $\{\hat{\alpha}_1, \hat{\beta}_1\} = \{-0.8132, 0.0223\}$ obtained from the Step-1 regression and consider other exogenous factors in Step 2 to improve the predictive power.

Step-2 Regression Results with Linear Model. With annual county-level data for GDPs and education levels, we obtain one data point for each county in each year. Specifically, for county $j \in J$ in each year, the response variable is given by $\widetilde{d}_j = d_j(\boldsymbol{z}, \boldsymbol{w}) - \bar{d}_j - \sum_{i \in I} \min\{t_i, T\} e^{\hat{\alpha}_1 - \hat{\beta}_1 \text{dist}_{ij}} z_i$. Here, $\bar{d}_j$ represents the EV registration at the beginning of that year, and $d_j(\boldsymbol{z}, \boldsymbol{w})$ represents the EV registration at the end of that year, influenced by the charger capacity decisions $\boldsymbol{z}$ and other factors $\boldsymbol{w}$, and the term $\sum_{i \in I} \min\{t_i, T\} e^{\hat{\alpha}_1 - \hat{\beta}_1 \text{dist}_{ij}} z_i$ measures the total demand contribution by endogenous factors (e.g., EV charging stations built within the corresponding year). After grouping data points among all counties from 2017 to 2020, we construct a Step-2 training dataset $\mathcal{D}_2 = \{\mathrm{g}^k, \mathrm{e}^k, \widetilde{d}^k\}_{k=1}^{N_2}$ with $N_2 = 219$ and solve the optimization model (14) via OLS to obtain the best parameters $(\hat{\alpha}_2, \hat{\beta}_2, \hat{\beta}_3)$ for the linear regression model in Eq. (9).

On the other hand, because the decision-independent regression model (25) does not contain the endogenous component, we do not need to perform Step-1 regression. In Step 2, the response variable is instead given by $\widetilde{d}_j^{DI} := d_j^{DI}(\mathbf{g}, \mathbf{e}) - \bar{d}_j$, where $d_j^{DI}(\mathbf{g}, \mathbf{e})$ denotes the EV registration at the end of the corresponding year, influenced solely by the exogenous factors $(\mathbf{g}, \mathbf{e})$. Note that both $d_j(\boldsymbol{z}, \boldsymbol{w})$ and $d_j^{DI}(\mathbf{g}, \mathbf{e})$ share the same data. Similarly, we construct a training dataset $\mathcal{D}_2 = \{\mathrm{g}^k, \mathrm{e}^k, \widetilde{d}^{k,DI}\}_{k=1}^{N_2}$ with $N_2 = 219$ and regress a linear model using OLS by solving the optimization model (14) for the decision-independent counterpart (25) to get $(\hat{\alpha}_2^{DI}, \hat{\beta}_2^{DI}, \hat{\beta}_3^{DI})$.

We use the annual data for GDPs and education levels (both normalized to $[0, 1]$) from 2017 to 2020 as the training dataset with a size of 219, and from 2021 to 2022 as the testing dataset with a size of 109. We adopt `LinearRegression` from scikit-learn library to conduct the regression analysis. The regression results are summarized in Table 6, where we present the coefficients and intercepts in the linear regression model, as well as the MSE, MAE, ME, and R^2 in the training and

testing datasets for decision-dependent and decision-independent models, respectively. We mark the best results in bold.

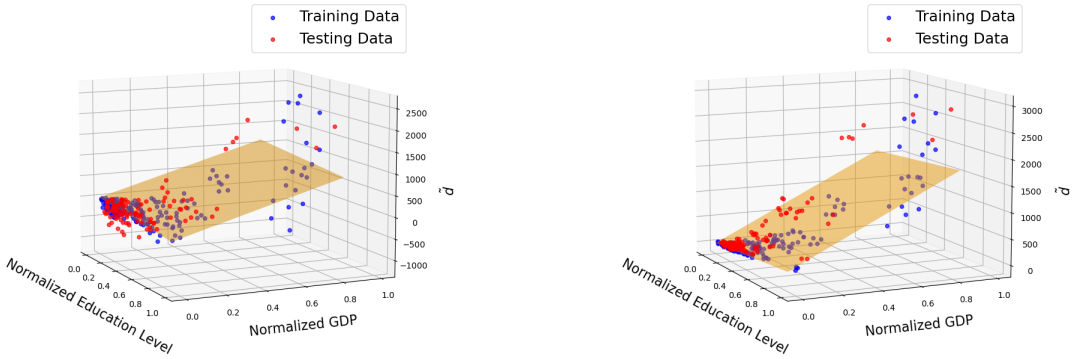
Table 6: Step-2 regression comparison between decision-dependent and decision-independent models with linear regression.

	Metrics	Decision-Dependent	Decision-Independent
Coefficients for Linear Regression	Coefficient (GDP)	1,191.0565	1,773.5902
	Coefficient (Education Level)	192.0270	373.5067
	Intercept	-192.3205	-83.8339
Training (2017–2020)	MSE	127,356.4024	**51,489.8911**
	MAE	117.4855	**39.9879**
	ME	1,611.2392	**1,361.1660**
	R^2	0.4126	**0.8020**
Testing (2021–2022)	MSE	**114,428.0481**	132,013.7640
	MAE	154.0327	**79.1276**
	ME	1,433.8794	**1,425.6726**
	R^2	0.5361	**0.6395**

Table 6 reveals that GDP is the dominant factor influencing customer demand, exhibiting a magnitude larger coefficient than education level in both model formulations. Both exogenous factors demonstrate a positive correlation with demand. The decision-dependent model achieves a lower MSE in the testing dataset, while the decision-independent counterpart obtains a lower error in all other metrics.

Figure 9 provides 3-dimensional scatter plots of the Step-2 regression results for both decision-dependent and decision-independent models. In Figure 9a, the z-axis measures the response variable $\widetilde{d}^k$ in the decision-dependent model, while the x- and y-axes represent the normalized education level and normalized GDP, respectively. In contrast, Figure 9b use the same x- and y-axes but plot a different response variable $\widetilde{d}^{k,DI}$ in the decision-independent model on the z-axis. In all figures, the darker blue dots correspond to the training data points, the lighter red dots correspond to the testing data points, and the orange surface illustrates the fitted linear regression model. From the figure, the relationship between the response variable and features is not exactly linear, which motivates us to utilize nonparametric regression methods in subsequent analyses.

Step-2 Regression with Nonparametric Model. In this section, we extend the analyses to nonparametric regression models. Given the decision-dependent term in our regression model (9), applying nonparametric regression methods to the training dataset directly presents significant challenges. Consequently, we follow the two-step regression approach and fix the results for Step-1 as $\{\hat{\alpha}_1, \hat{\beta}_1\} = \{-0.8132, 0.0223\}$. With the same training dataset $\mathcal{D}_2$, we adopt kNN, decision trees, random forest, and gradient boosting methods to conduct the regression analysis, using `KNeighborsRegressor`, `DecisionTreeRegressor`, `RandomForestRegressor`, and `HistGradientBoostingRegressor` in the scikit-learn library, respectively. We utilize the `GridSearchCV` method from the scikit-learn library to conduct 5-fold cross-validation hyperparameter tuning. This



(a) Response $\widetilde{d}^k$ v.s. Normalized Features $(\mathrm{g}^k, \mathrm{e}^k)$ for Decision-Dependent Regression Model

(b) Response $\widetilde{d}^{k,DI}$ v.s. Normalized Features $(\mathrm{g}^k, \mathrm{e}^k)$ for Decision-Independent Regression Model

Figure 9: Response vs. normalized features for Step-2 regression with linear models.

involves an exhaustive search over a specified range of parameter values for each parameter in each model, systematically evaluating different combinations to determine the optimal configuration that minimizes prediction error. After completing the hyperparameter tuning process using `GridSearchCV`, the parameters selected for each model are presented in Table 12.

Under these fine-tuned hyperparameters, we conduct our analysis and summarize the performance of different regression models in Table 7, including MSE, MAE, and ME values for the training and testing datasets, where we mark the best performance in bold. We do not report the R^2 scores in Table 7 because it is not a good metric for nonparametric regression models.

Table 7: Step-2 regression comparison between decision-dependent and decision-independent models with nonparametric regression.

Regression Model	Decision Dependency	Training			Testing		
		MSE	MAE	ME	MSE	MAE	ME
kNN	Decision-Dependent	102,348.54	90.74	1,754.47	**123,476.34**	149.87	**1,512.72**
	Decision-Independent	**46,917.63**	**16.77**	**1,301.44**	157,200.44	**79.22**	1,604.44
Decision Tree	Decision-Dependent	72,964.74	89.48	1,644.57	**104,047.09**	161.85	**1,070.61**
	Decision-Independent	**16,533.49**	**19.90**	**1,000.57**	201,157.20	**79.09**	1,659.00
Random Forest	Decision-Dependent	77,348.37	91.49	1,700.95	**112,824.01**	164.91	**1,275.39**
	Decision-Independent	**30,564.10**	**22.19**	**968.04**	150,283.80	**76.15**	1,383.65
Gradient Boosting	Decision-Dependent	126,335.18	78.46	1,949.97	151,469.07	169.95	1,445.52
	Decision-Independent	**39,301.48**	**13.49**	**1,188.97**	**122,903.43**	**71.40**	**1,245.36**

Table 7 indicates that the decision-independent counterpart achieves better in-sample performance over decision-dependent regression model, exhibiting lower MSE, MAE and ME across all nonparametric methods in the training dataset. However, decision-dependent regression model achieves better out-of-sample predictive power with lower MSE and ME for all nonparametric methods except for Gradient Boosting. The decision-dependent regression model also yields smaller performance gaps between training and testing datasets compared to the decision-independent approach, suggesting more robust performance across different datasets.

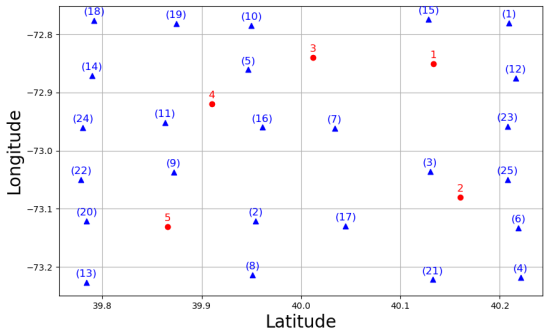


Figure 10: Locations of customer sites and potential EV charging stations used in downstream optimization problem.

5.2 Downstream Optimization Results

After obtaining the prediction models and empirical residuals, we next solve our downstream ER-DD-SAA problem (17) to optimally locate EV charging stations. In this section, we consider a synthetic optimization problem setup to compare the in-sample performance between our ER-DD-SAA model and its decision-independent counterpart (ER-DI-SAA). Note that we cannot run downstream optimization models based on real-world data because the ground truth demand model remains unknown and it is hard to compare the performance between our solutions and real deployment decisions in the decision-dependent setting.

5.2.1 Parameter Setup

We randomly generate a set of potential locations for EV charging stations $i \in I$ and a set of customer sites to be served by these stations $j \in J$ over a geographical area centered at the coordinate $(40, -73)$, within a 15-mile radius (see Figure 10). For this analysis, we focus on a setting in which we consider $I = 25$ potential EV charging station locations with a maximum number of chargers $M_i = 10$ for each location $i \in I$, and we have $J = 5$ customer sites. The locations of candidate EV charging stations and customer sites are illustrated in Figure 10. The parameter setup is the same as Section 4.3, except that we set the unit revenue r_j to 20 and the unit penalty p_j for unmet demand to 10. The initial demand at each customer site $\bar{d}_j$ is set to 15,000. We set the GDPs and education levels for all customer sites to 3,853,137 and 26.7%, which are the median values among all counties in New York State in the year 2017 (Federal Reserve Bank of St. Louis, b,a) and are normalized to 0.3 and 0.2, respectively. Additionally, we assume $T = 30$ and $\min\{t_i, T\} = T$ to simplify the downstream optimization model, as discussed in Section 3.4.

5.2.2 Result Analyses

We compare the in-sample performance between ER-DD-SAA and ER-DI-SAA across different regression models in Table 8, where we present the optimal first-stage decisions $\boldsymbol{y}$ and $\boldsymbol{z}$, the optimal total number of chargers, and the optimal in-sample cost. We mark the better results in bold and

the differences in the location decisions with *. The detailed in-sample cost breakdown is displayed in Table 9, where again the better cost/revenue values are marked in bold. From Table 8, the ER-DD-SAA model recommends installing more EV chargers compared to its decision-independent counterpart, ultimately resulting in better in-sample optimal objective values. The cost breakdown presented in Table 9 further reveals that building more chargers in ER-DD-SAA increases the fixed cost, charger installment cost, and transportation cost, but reduces the penalty cost and brings significantly more revenue due to the increased demand according to (9). Moreover, both models open 10 chargers (the maximum amount) at locations around the center (e.g., locations #2, 3, 7, 9, 11, 16), which can reduce the total transportation cost. While ER-DI-SAA chooses to install chargers at locations #6 and #20, ER-DD-SAA chooses locations #17 and #23 instead that are closer to most customer sites and thus can increase the demand most according to the distance effect in (9).

Table 8: Optimal first-stage solutions and optimal objective values using different regression models.

Regression	Optimization	Location and Capacity Decision	# Chargers	In-Sample Cost
Linear Model	ER-DD-SAA	[22]: 4 chargers [24*]: 7 chargers [1, 2, 3, 5, 7, 8*, 9, 10, 11, 12, 15, 16, 17*, 19, 23*, 25]: 10 chargers	**171**	**-1,468,359.74**
	ER-DI-SAA	[21*]: 2 chargers [22]: 5 chargers [1, 2, 3, 5, 6*, 7, 9, 10, 11, 12, 15, 16, 19, 20*, 25]: 10 chargers	157	-1,349,757.26
kNN	ER-DD-SAA	[22*]: 3 chargers [24*]: 6 chargers [1, 2, 3, 5, 7, 8*, 9, 10, 11, 12, 15, 16, 17*, 19, 23, 25]: 10 chargers	**169**	**-1,419,887.71**
	ER-DI-SAA	[23]: 1 charger [1, 2, 3, 5, 6*, 7, 9, 10, 11, 12, 15, 16, 19, 20*, 25]: 10 chargers	151	-1,324,382.38
Decision Tree	ER-DD-SAA	[24*]: 2 chargers [1, 2, 3, 5, 7, 8*, 9, 10, 11, 12, 15, 16, 17*, 19, 23*, 25]: 10 chargers	**162**	**-1,383,501.17**
	ER-DI-SAA	[1, 2, 3, 5, 6*, 7, 9, 10, 11, 12, 15, 16, 19, 20*, 25]: 10 chargers	150	-1,325,630.09
Random Forest	ER-DD-SAA	[22*]: 3 chargers [24*]: 5 chargers [1, 2, 3, 5, 7, 8*, 9, 10, 11, 12, 15, 16, 17*, 19, 23*, 25]: 10 chargers	**168**	**-1,414,661.37**
	ER-DI-SAA	[1, 2, 3, 5, 6*, 7, 9, 10, 11, 12, 15, 16, 19, 20*, 25]: 10 chargers	150	-1,329,297.64
Gradient Boosting	ER-DD-SAA	[24*]: 6 chargers [1, 2, 3, 5, 7, 8*, 9, 10, 11, 12, 15, 16, 17*, 19, 23*, 25]: 10 chargers	**166**	**-1,440,976.81**
	ER-DI-SAA	[1, 2, 3, 5, 6*, 7, 9, 10, 11, 12, 15, 16, 19, 20*, 25]: 10 chargers	150	-1,324,446.45

Table 9: Cost-breakdown for in-sample experiments.

Regression	Optimization	Fixed Cost	Charger Cost	Trans. Cost	Penalty Cost	Revenue
Linear Model	ER-DD-SAA	144,000	68,400	2,434.38	4,931.11	**1,688,125.25**
	ER-DI-SAA	**136,000**	**62,800**	**2,039.15**	**1,698.02**	1,552,294.44
kNN	ER-DD-SAA	144,000	67,600	2,329.05	**1,929.79**	**1,635,746.57**
	ER-DI-SAA	**128,000**	**60,400**	**1,934.41**	7,216.28	1,521,933.07
Decision Tree	ER-DD-SAA	136,000	64,800	2,248.97	**1,492.39**	**1,588,042.54**
	ER-DI-SAA	**120,000**	**60,000**	**1,917.30**	8,402.60	1,515,950.00
Random Forest	ER-DD-SAA	144,000	67,200	2,318.75	**1,778.82**	**1,629,958.96**
	ER-DI-SAA	**120,000**	**60,000**	**1,916.52**	12,388.77	1,523,602.94
Gradient Boosting	ER-DD-SAA	136,000	66,400	2,372.88	**5,003.00**	**1,650,752.70**
	ER-DI-SAA	**120,000**	**60,000**	**1,915.57**	8,250.75	1,514,612.79

Tables 8 and 9 reveal a superior in-sample performance of our ER-DD-SAA model, compared to ER-DI-SAA, in all regression setups. This highlights the importance of considering decision dependency in the EV charging station location problem. However, since the ground truth demand model is unknown and it is expensive to run simulations in the field (i.e., invest EV chargers at corresponding optimal locations and then observe the actual demand in the real world), it remains challenging to compare the out-of-sample performance of the two models.

5.3 Discussion

Let us now look at the New York State results conducted in this section in light of the synthetic experiments conducted in Section 4. It appears that, in the real-world dataset from New York State, there is mild decision dependency. Therefore, our regression results show modest success with decision-dependent regression (see Tables 6 and 7), compared to the synthetic experiments with higher α_1 values (see Table 3). At the same time, they highlight the importance of considering contextual information for accurate demand prediction. These results also imply that even with moderate decision dependency, incorporating such dependency into the optimization model yields superior results, increasing overall revenue more than the increased costs of opening additional facilities (see Tables 8 and 9 from the case study and Table 4 from the synthetic experiments). Consequently, these results indicate the necessity and superiority of considering decision-dependent contextual stochastic optimization for the EV charging station location problem.

6 Conclusions

In this paper, we studied a contextual stochastic EV charging station location problem, where customer demand could be affected by both endogenous factors such as location/capacity decisions and exogenous factors such as GDPs and education levels. To incorporate the estimation error in learning the demand, we extended the ER-SAA approach (Kannan et al., 2025) to the decision-dependent setting, namely ER-DD-SAA, and theoretically analyzed its consistency and asymptotic

optimality. We utilized a nonlinear regression function to describe the relationship between the customer demand and these endogenous/exogenous covariates and proposed a two-step regression approach, where we first regressed on the endogenous capacity decision and then regressed on the other exogenous factors. To better conduct out-of-sample tests, we designed synthetic datasets with a known ground truth demand model. Under most settings, our decision-dependent model achieves better predictive performance and smaller out-of-sample costs compared to the decision-independent counterpart. Moreover, as we increase the degree of decision dependency in the ground truth model, the gap between our ER-DD-SAA model and the decision-independent counterpart becomes larger. Through a case study based on real-world data in New York State, our regression results indicate that the decision-dependent regression model exhibits better out-of-sample predictive power compared to the decision-independent counterpart. Accordingly, the ER-DD-SAA model achieves a better in-sample performance compared to the decision-independent counterpart.

There are several directions that are worth investigating. This study considered a two-stage EV charging station location problem, where we did not optimize the sequence and timing of building EV charging stations. It will be interesting to consider a sequential decision-making problem that incorporates dynamic demand learning. While this paper utilized linear regression models and nonparametric regression approaches (e.g., kNN, decision tree, random forest, and gradient boosting), other machine learning techniques such as (deep) neural networks can also be employed here, which we leave as future work.

Acknowledgments

This work has been supported in part by the United States National Science Foundation [grant number 2331782] of Xian Yu and by the U.S. Department of Energy, Office of Science, Office of Advanced Scientific Computing Research under Grant DE-SC0023361 of Güzin Bayraksan.

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A Details of All Omitted Proofs

Proof of Proposition 1. Since the feasible set $\mathcal{H}(\boldsymbol{z}, \boldsymbol{w})$ for the second-stage problem $h(\boldsymbol{z}, \boldsymbol{d}(\boldsymbol{z}, \boldsymbol{w}))$ is always non-empty and bounded, strong duality holds and we have

$$\begin{aligned} h(\boldsymbol{z}, \boldsymbol{d}(\boldsymbol{z}, \boldsymbol{w})) = \max_{\boldsymbol{u}, \boldsymbol{\gamma}} \ & (\boldsymbol{u} - \boldsymbol{r})^{\mathsf{T}} \boldsymbol{d}(\boldsymbol{z}, \boldsymbol{w}) - \sum_{i \in I} \gamma_i C_i z_i \\ \text{s.t. } & u_j - \gamma_i \leq c_{ij}, \ \forall i \in I, \ j \in J \\ & u_j \leq p_j, \ \forall j \in J \\ & \gamma_i \geq 0, \ \forall i \in I \end{aligned}$$

where $\boldsymbol{u} \in \mathbb{R}^{|J|}$ and $\boldsymbol{\gamma} \in \mathbb{R}^{|I|}$ denote the dual variables associated with the constraints of the second-stage problem. We denote the feasible region of the dual problem as

$$\mathcal{F} = \left\{ \boldsymbol{u} \in \mathbb{R}^{|J|}, \boldsymbol{\gamma} \in \mathbb{R}^{|I|} \ \middle| \ u_j - \gamma_i \leq c_{ij}, \ \forall i \in I, j \in J, \ u_j \leq p_j, \ \forall j \in J, \ \gamma_i \geq 0, \ \forall i \in I \right\}.$$

Let $\text{vert}(\mathcal{F})$ denote the set of extreme points of the dual feasible set $\mathcal{F}$. Since $\mathcal{F}$ is defined by finitely many linear inequalities, $\text{vert}(\mathcal{F})$ is finite. Then we have

$$\begin{aligned} & \left| c(\boldsymbol{y}, \boldsymbol{z}; \bar{\boldsymbol{d}}(\boldsymbol{z}, \boldsymbol{w})) - c(\boldsymbol{y}, \boldsymbol{z}; \boldsymbol{d}(\boldsymbol{z}, \boldsymbol{w})) \right| \\ = & \left| \max_{\boldsymbol{u}, \boldsymbol{\gamma} \in \text{vert}(\mathcal{F})} \left\{ (\boldsymbol{u} - \boldsymbol{r})^{\mathsf{T}} \bar{\boldsymbol{d}}(\boldsymbol{z}, \boldsymbol{w}) - \sum_{i \in I} \gamma_i C_i z_i \right\} - \max_{\boldsymbol{u}, \boldsymbol{\gamma} \in \text{vert}(\mathcal{F})} \left\{ (\boldsymbol{u} - \boldsymbol{r})^{\mathsf{T}} \boldsymbol{d}(\boldsymbol{z}, \boldsymbol{w}) - \sum_{i \in I} \gamma_i C_i z_i \right\} \right| \\ \leq & \max_{\boldsymbol{u}, \boldsymbol{\gamma} \in \text{vert}(\mathcal{F})} \left| \left((\boldsymbol{u} - \boldsymbol{r})^{\mathsf{T}} \bar{\boldsymbol{d}}(\boldsymbol{z}, \boldsymbol{w}) - \sum_{i \in I} \gamma_i C_i z_i \right) - \left((\boldsymbol{u} - \boldsymbol{r})^{\mathsf{T}} \boldsymbol{d}(\boldsymbol{z}, \boldsymbol{w}) - \sum_{i \in I} \gamma_i C_i z_i \right) \right| \\ = & \max_{\boldsymbol{u}, \boldsymbol{\gamma} \in \text{vert}(\mathcal{F})} \left| (\boldsymbol{u} - \boldsymbol{r})^{\mathsf{T}} \left(\bar{\boldsymbol{d}}(\boldsymbol{z}, \boldsymbol{w}) - \boldsymbol{d}(\boldsymbol{z}, \boldsymbol{w}) \right) \right| \\ \overset{(a)}{\leq} & \| \bar{\boldsymbol{d}}(\boldsymbol{z}, \boldsymbol{w}) - \boldsymbol{d}(\boldsymbol{z}, \boldsymbol{w}) \| \max_{\boldsymbol{u}, \boldsymbol{\gamma} \in \text{vert}(\mathcal{F})} \| \boldsymbol{u} - \boldsymbol{r} \| \end{aligned}$$

where (a) uses Cauchy-Schwarz inequality. Since the set $\text{vert}(\mathcal{F})$ is finite, the Lipschitz constant $L = \max_{\boldsymbol{u}, \boldsymbol{\gamma} \in \text{vert}(\mathcal{F})} \| \boldsymbol{u} - \boldsymbol{r} \| < +\infty$.

□

Before proving Theorem 1, let us first prove some useful lemmas. We define the difference between the i-th scenario for FI-DD-SAA and the i-th scenario for ER-DD-SAA as

$$\tilde{\boldsymbol{\epsilon}}_n^i(\boldsymbol{z}, \boldsymbol{w}) := \left(\hat{Q}_n(\boldsymbol{z}, \boldsymbol{w}) + \hat{\boldsymbol{\epsilon}}^i \right) - \left(Q^*(\boldsymbol{z}, \boldsymbol{w}) + \boldsymbol{\epsilon}^i \right), \ \forall i \in [n]. \tag{26}$$

Lemma 1. Based on Proposition 1, for each $\boldsymbol{w} \in \mathcal{W}$, we have

$$\sup_{\boldsymbol{y}, \boldsymbol{z} \in \mathcal{Z}} |\hat{g}_n^{ER}(\boldsymbol{y}, \boldsymbol{z}; \boldsymbol{w}) - g_n^*(\boldsymbol{y}, \boldsymbol{z}; \boldsymbol{w})| \leq \sup_{\boldsymbol{z} \in \mathcal{Z}_z} \frac{1}{n} \sum_{i=1}^n L \| \tilde{\boldsymbol{\epsilon}}_n^i(\boldsymbol{z}, \boldsymbol{w}) \|$$

Proof. For each $\boldsymbol{w} \in \mathcal{W}$,

$$\begin{aligned}
\sup_{\boldsymbol{y},\boldsymbol{z}\in\mathcal{Z}} |\hat{g}_n^{ER}(\boldsymbol{y},\boldsymbol{z};\boldsymbol{w}) - g_n^*(\boldsymbol{y},\boldsymbol{z};\boldsymbol{w})| &= \sup_{\boldsymbol{y},\boldsymbol{z}\in\mathcal{Z}} \Big| \frac{1}{n}\sum_{i=1}^{n} c(\boldsymbol{y},\boldsymbol{z};\hat{Q}_n(\boldsymbol{z},\boldsymbol{w}) + \hat{\boldsymbol{\epsilon}}^i) - \frac{1}{n}\sum_{i=1}^{n} c(\boldsymbol{y},\boldsymbol{z};Q^*(\boldsymbol{z},\boldsymbol{w}) + \boldsymbol{\epsilon}^i) \Big| \\
&\leq \sup_{\boldsymbol{y},\boldsymbol{z}\in\mathcal{Z}} \frac{1}{n}\sum_{i=1}^{n} \Big| c(\boldsymbol{y},\boldsymbol{z};\hat{Q}_n(\boldsymbol{z},\boldsymbol{w}) + \hat{\boldsymbol{\epsilon}}^i) - c(\boldsymbol{y},\boldsymbol{z};Q^*(\boldsymbol{z},\boldsymbol{w}) + \boldsymbol{\epsilon}^i) \Big| \\
&\overset{(a)}{\leq} \sup_{\boldsymbol{z}\in\mathcal{Z}_z} L\Big(\frac{1}{n}\sum_{i=1}^{n} \|\tilde{\boldsymbol{\epsilon}}_n^i(\boldsymbol{z},\boldsymbol{w})\| \Big)
\end{aligned}$$

where (a) is due to Proposition 1. □

Lemma 2. Given the estimated regression function $\hat{Q}_n(\boldsymbol{z},\boldsymbol{w})$, for each $\boldsymbol{w} \in \mathcal{W}, \boldsymbol{z} \in \mathcal{Z}_z$, we have

$$\frac{1}{n}\sum_{i=1}^{n} \|\tilde{\boldsymbol{\epsilon}}_n^i(\boldsymbol{z},\boldsymbol{w})\| \leq \|\hat{Q}_n(\boldsymbol{z},\boldsymbol{w}) - Q^*(\boldsymbol{z},\boldsymbol{w})\| + \frac{1}{n}\sum_{i=1}^{n} \|\hat{Q}_n(\boldsymbol{z}^i,\boldsymbol{w}^i) - Q^*(\boldsymbol{z}^i,\boldsymbol{w}^i)\|$$

Proof.

$$\begin{aligned}
\frac{1}{n}\sum_{i=1}^{n} \|\tilde{\boldsymbol{\epsilon}}_n^i(\boldsymbol{z},\boldsymbol{w})\| &= \frac{1}{n}\sum_{i=1}^{n} \|\big(\hat{Q}_n(\boldsymbol{z},\boldsymbol{w}) + \hat{\boldsymbol{\epsilon}}^i\big) - \big(Q^*(\boldsymbol{z},\boldsymbol{w}) + \boldsymbol{\epsilon}^i\big)\| \\
&\leq \|\hat{Q}_n(\boldsymbol{z},\boldsymbol{w}) - Q^*(\boldsymbol{z},\boldsymbol{w})\| + \frac{1}{n}\sum_{i=1}^{n} \|\hat{\boldsymbol{\epsilon}}^i - \boldsymbol{\epsilon}^i\| \\
&= \|\hat{Q}_n(\boldsymbol{z},\boldsymbol{w}) - Q^*(\boldsymbol{z},\boldsymbol{w})\| + \frac{1}{n}\sum_{i=1}^{n} \|\hat{Q}_n(\boldsymbol{z}^i,\boldsymbol{w}^i) - Q^*(\boldsymbol{z}^i,\boldsymbol{w}^i)\|
\end{aligned}$$

□

Lemma 3. Suppose Assumption 2 holds. Then, for a.e. $\boldsymbol{w} \in \mathcal{W}$, the sequence of objective functions of the ER-DD-SAA (8) converges in probability to the objective function of the true problem (6) uniformly on the feasible region $\mathcal{Z}$.

Proof of Lemma 3. The difference between the objective functions of the ER-DD-SAA problem (8) and true problem (6) can be bounded uniformly over decision variables $\boldsymbol{y},\boldsymbol{z} \in \mathcal{Z}$ as follows:

$$\sup_{\boldsymbol{y},\boldsymbol{z}\in\mathcal{Z}} |\hat{g}_n^{ER}(\boldsymbol{y},\boldsymbol{z};\boldsymbol{w}) - g(\boldsymbol{y},\boldsymbol{z};\boldsymbol{w})| \leq \sup_{\boldsymbol{y},\boldsymbol{z}\in\mathcal{Z}} |\hat{g}_n^{ER}(\boldsymbol{y},\boldsymbol{z};\boldsymbol{w}) - g_n^*(\boldsymbol{y},\boldsymbol{z};\boldsymbol{w})| + \sup_{\boldsymbol{y},\boldsymbol{z}\in\mathcal{Z}} |g_n^*(\boldsymbol{y},\boldsymbol{z};\boldsymbol{w}) - g(\boldsymbol{y},\boldsymbol{z};\boldsymbol{w})|$$

The second term on the r.h.s of the above inequality vanishes in the limit in probability by Proposition 2. By Assumption 2, $\frac{1}{n}\sum_{i=1}^{n} \|\tilde{\boldsymbol{\epsilon}}_n^i(\boldsymbol{z},\boldsymbol{w})\|$ converges to zero uniformly on $\mathcal{Z}_z$ in probability, and thus the first term on the r.h.s. converges to zero in probability according to Lemmas 1 and 2. By $o_p(1) + o_p(1) = o_p(1)$, $\sup_{\boldsymbol{y},\boldsymbol{z}\in\mathcal{Z}} |\hat{g}_n^{ER}(\boldsymbol{y},\boldsymbol{z};\boldsymbol{w}) - g(\boldsymbol{y},\boldsymbol{z};\boldsymbol{w})|$ converges to zero in probability for a.e. $\boldsymbol{w} \in \mathcal{W}$. □

Proof of Theorem 1. Let $(\boldsymbol{y}^*(\boldsymbol{w}), \boldsymbol{z}^*(\boldsymbol{w})) \in S^*(\boldsymbol{w})$ and $(\hat{\boldsymbol{y}}_n^{ER}(\boldsymbol{w}), \hat{\boldsymbol{z}}_n^{ER}(\boldsymbol{w})) \in \hat{S}_n^{ER}$. Consider any constant $\delta > 0$. From Lemma 3, we have for a.e. $\boldsymbol{w} \in \mathcal{W}$:

$$P\{|\hat{g}_n^{ER}(\hat{\boldsymbol{y}}_n^{ER}, \hat{\boldsymbol{z}}_n^{ER}; \boldsymbol{w}) - v^*(\boldsymbol{w})| > \delta\} \to 0 \implies P\{\hat{v}_n^{ER} > v^*(\boldsymbol{w}) + \delta\} \to 0 \quad \text{and}$$
$$P\{|\hat{v}_n^{ER}(\boldsymbol{w}) - g(\hat{\boldsymbol{y}}_n^{ER}, \hat{\boldsymbol{z}}_n^{ER}; \boldsymbol{w})| > \delta\} \to 0 \implies P\{v^*(\boldsymbol{w}) > \hat{v}_n^{ER} + \delta\} \to 0$$

As a result, we have $P\{|v^*(\boldsymbol{w}) - \hat{v}_n^{ER}| > \delta\} \to 0$ and thus $\hat{v}_n^{ER} \xrightarrow{P} v^*(\boldsymbol{w})$ for a.e. $\boldsymbol{w} \in \mathcal{W}$. The rest of the proof follows the proof of Theorem 5 in Kannan et al. (2025). □

B Detailed Parameter Setup

Table 10: A snapshot of EV registration dataset by county in New York State.

Year	Month	Technology	EV Original Registrations	County
2017	January	BEV	3	Ulster County
2017	January	BEV	1	Warren County
2017	January	PHEV	3	Bronx County
2017	January	PHEV	2	Broome County

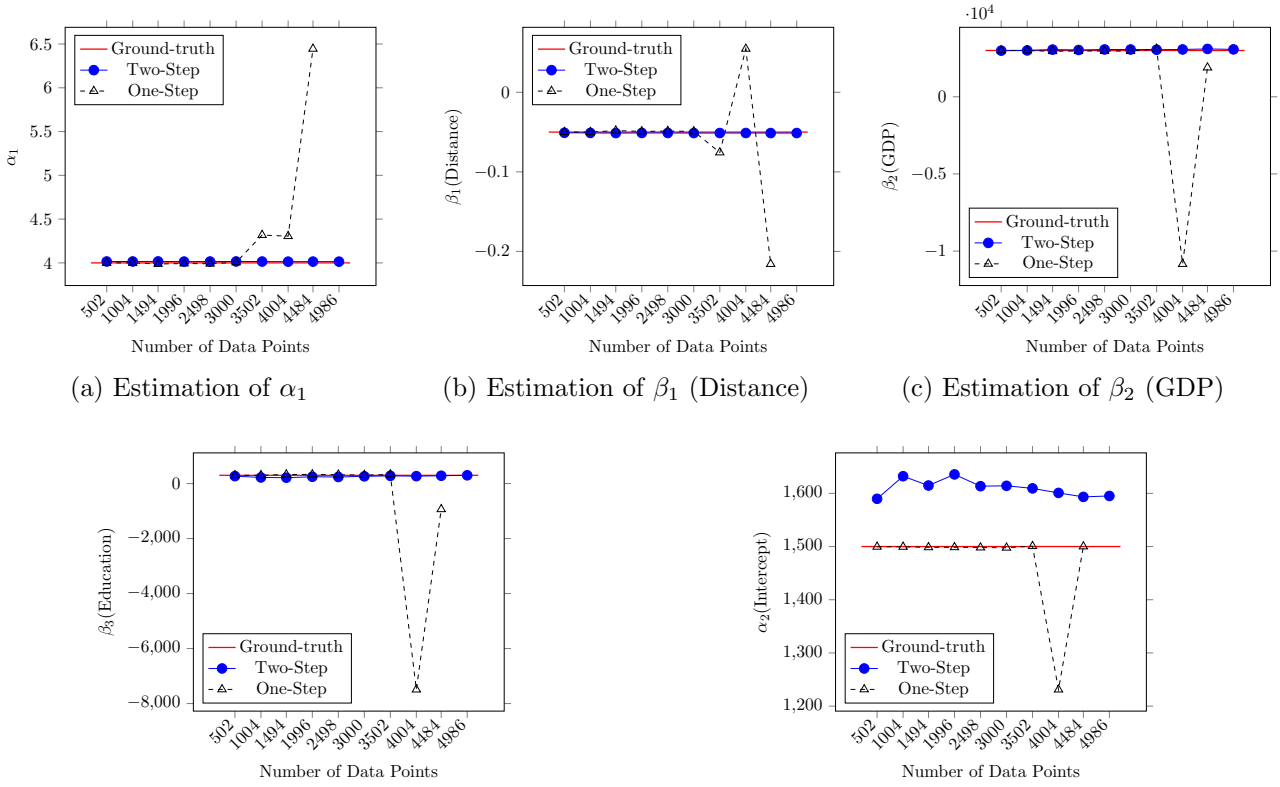
Table 11: A snapshot of EV charging station dataset in New York State.

# Publicly Accessible EV Chargers	Latitude	Longitude	Open Date
3	40.755072	-73.892094	3/1/17
2	43.053474	-76.163547	3/1/17
2	43.181259	-76.256973	4/1/17
3	41.980335	-73.990256	6/1/17

Table 12: Best hyperparameters in different nonparametric regression models for decision-dependent and decision-independent models.

Regression Model	Decision Dependency	Best Parameters
kNN	Decision-Independent	n_neighbors=9
	Decision-Dependent	n_neighbors=9
Decision Tree	Decision-Independent	max_depth=4, min_samples_split=8, random_state=40
	Decision-Dependent	max_depth=3, min_samples_split=5, random_state=40
Random Forest	Decision-Independent	max_depth=3, min_samples_split=9, n_estimators=50, random_state=42
	Decision-Dependent	max_depth=3, min_samples_split=9, n_estimators=50, random_state=45
Gradient Boosting	Decision-Independent	learning_rate=0.1, max_depth=3
	Decision-Dependent	learning_rate=0.01, max_depth=6

C Performances of Different Regression Approaches



(a) Estimation of α_1 (b) Estimation of β_1 (Distance) (c) Estimation of β_2 (GDP)

(d) Estimation of β_3 (Education) (e) Estimation of α_2 (Intercept)

Figure 11: Comparison of parameter estimation performance: one-step regression vs. two-step regression approaches